CAMPUS REUSABLE EXCHANGE NETWORK

A CIRCULAR CAMPUS SMART REUSABLE ITEM MANAGEMENT SYSTEM

A PROJECT REPORT

Submitted by

GOWTHAM D

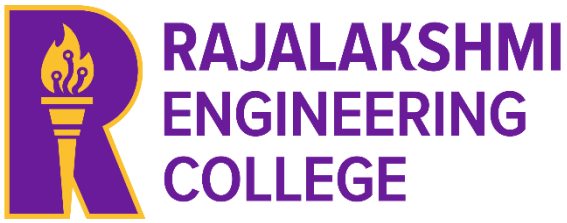
2116240701154

in partial fulfilment of the End Semester Examination for the course

CB23F35 –
LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING IN COMPUTER SCIENCE ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled **"CAMPUS REUSABLE EXCHANGE NETWORK"** is the bonafide work of GOWTHAM D (2116240701154) , who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026 – 2027 , Semester V , under my supervision.

SIGNATURE

Dr. Manoranjini J
HEAD OF THE DEPARTMENT
Department of Computer Science Engineering
Rajalakshmi Engineering College
Thandalam – 602 105

SIGNATURE

Ms. Aswana Lal
COURSE FACULTY - Assistant Professor
Department of Computer Science Engineering
Rajalakshmi Engineering College
Thandalam – 602 105

Submitted for the End Semester Examination (Individual Project) held on _____

INTERNAL EXAMINER

EXTERNAL EXAMINER

DECLARATION

I, GOWTHAM D (2116240701154), hereby declare that the project report titled “CAMPUS REUSABLE EXCHANGE NETWORK”, submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia AI fields and the AI agent), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai **SIGNATURE OF THE STUDENT**

Date: GOWTHAM D

2116240701154

ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman Thiru. S. Meganathan , Chairperson Dr. (Mrs.) Thangam Meganathan and Vice-Chairman Mr. Abhay Shankar Meganathan for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Dr. S.N. Murugesan , Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank Dr. Manoranjini J , Head of the Department, Department of Computer Science Engineering , for the guidance and support provided throughout the course.

I owe my sincere thanks to my course faculty, Ms. Aswana Lal , Assistant Professor , for the valuable guidance, reviews and feedback that shaped this application and report.

I also acknowledge Zoho Corporation / Zoho Creator Education Programme for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

GOWTHAM D

ABSTRACT

Educational campuses buy, use and discard large quantities of laboratory coats, equipment, chargers, furniture and other items, while identical items sit unused elsewhere on the same campus. Exchange of such items is informal, relying on messaging groups, phone calls and paper registers, so availability is unknown, approvals are unrecorded, overdue items go unnoticed and damaged items are re-issued. This project addresses that problem for students, faculty, staff and administrators of a campus.

A low-code approach on Zoho Creator was chosen because the process is form-centric and workflow-driven, and because hosting, security, mobile access and AI features are built in. The Campus Reusable Exchange Network application contains nine forms (Campus_Users, Item_Categories, Reusable_Items, Reuse_Requests, Item_Loans, Item_Returns, Maintenance_Requests, Item_Transfers and Item_Donations) connected by lookup fields, list reports for each form and a Dashboard page with KPIs and charts. Automation is delivered through Deluge scripts and six form workflows, two scheduled workflows, an approval process for donations, a Blueprint for the delivery process, a batch workflow for loan imports and a cancellation function. Zia AI fields predict categories and keywords, verify images and recommend maintenance, an AI agent assists users in chat, and a Zoho Invoice connection supports item invoices.

The application was tested form by form, script by script and role by role, and prepared for publication with approval controls and an OAuth-based integration. The result is a single governed platform that gives visibility of items, an audit trail for requests and loans, automatic overdue and stock alerts and a live dashboard, replacing manual registers and chat-based coordination.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Circular Economy; Campus Reuse Management

PROJECT AT A GLANCE

Attribute	Details
Project Title	CAMPUS REUSABLE EXCHANGE NETWORK – A Circular Campus Smart Reusable Item Management System
Application Domain	Education / Sustainability – campus resource management
Student Name & Register No.	GOWTHAM D – 2116240701154
Zoho Creator Edition	Trial (at time of development) – 30 Days
Application Name (Link Name)	Campus Reusable Exchange Network (CRE)
Published / Live URL	https://gowthamdevendran1406.zohocreatorportal.in
Demo Credentials (per role)	[Admin: gowthamdevendran1406@gmailcom ▏ Staff: gowthamdevendhiran1406@gmail.com ▏ Student: haripraveen.n.2024.cse@rajalakshmi.edu.in , gowtham.d.2024.cse@rajalakshmi.edu.in]
Number of Forms / Subforms	9 / 2 (related-record sections)
Number of Reports (by type)	List: 4 ▏ Kanban: 3 ▏ Pivot/Chart: charts on Dashboard : 6 ▏Time line : 1 ▏ Spreadsheet : 1
Number of Pages / Dashboards	1 (Dashboard)
Number of Workflows	Form: 9 ▏ Schedule: 2 ▏ Approval: 1 ▏ Blueprint: 1 ▏ Batch: 1
Deluge Functions / Custom Actions	1 function (cancelOrder) + 2 workflow scripts
Integrations	Zoho Invoice
Roles & Profiles Defined	Administrator, Faculty, Staff, Student
Environments Used	Development → [Stage (1) → Production (1)]
Date of Production Deployment	31/10/2026

TABLE OF CONTENTS

ABSTRACT iv
PROJECT AT A GLANCE v
LIST OF TABLES viii
LIST OF FIGURES ix
LIST OF ABBREVIATIONS x
CHAPTER 1: INTRODUCTION 1
1.1 Background 1
1.2 Problem Statement 1
1.3 Objectives 2
1.4 Scope of the Project 2
1.5 Justification for a Low-Code Approach 2
1.6 Organisation of the Report 3
CHAPTER 2: LITERATURE SURVEY AND SYSTEM STUDY 4
2.1 Study of the Existing System 4
2.2 Limitations of the Existing System 4
2.3 Low-Code versus Traditional Development 4
2.4 Literature Survey 5
2.5 Emerging Trends in Low-Code Development 6
2.6 Proposed System 6
CHAPTER 3: REQUIREMENT ANALYSIS 7
3.1 Stakeholders and User Roles 7
3.2 Functional Requirements 7
3.3 Non-Functional Requirements 8
3.4 Hardware and Software Requirements 8
3.5 Use Case Model 9
CHAPTER 4: SYSTEM DESIGN 10
4.1 System Architecture 10
4.2 Data Model 10
4.3 Form Design 12
4.4 Process and Workflow Design 17
4.5 User Interface and Navigation Design 18
4.6 Access Control Design 18
CHAPTER 5: IMPLEMENTATION 19
5.1 Application Setup 19
5.2 Forms and Data Capture 19
5.3 Reports 29
5.4 Pages and Dashboards 30

5.5 Deluge Scripting 31
5.6 Workflows and Automation 36
5.7 Integrations 39
5.8 User Management and Role-Based Access 40
5.9 AI and Analytics Features 40
5.10 Pro-Code Extension 41
5.11 Mobile and Portal Access 41
CHAPTER 6: TESTING AND DEBUGGING 43
6.1 Testing Strategy 43
6.2 Test Cases 43
6.3 Debugging Deluge Scripts 44
6.4 Defect Log 44
6.5 User Acceptance Testing 44
CHAPTER 7: DEPLOYMENT, SECURITY AND GOVERNANCE 45
7.1 Environments and Deployment Strategy 45
7.2 Performance Optimisation 46
7.3 Security Features 46
7.4 Governance and Administration 47
CHAPTER 8: RESULTS AND DISCUSSION 48
8.1 Application Output 48
8.2 Achievement of Objectives 49
8.3 Comparison with the Existing System 50
8.4 Limitations of the Low-Code Approach 50
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum 50
CHAPTER 9: CONCLUSION AND FUTURE ENHANCEMENTS 51
9.1 Conclusion 51
9.2 Future Enhancements 51
9.3 Learning Outcomes and Career Relevance 51
REFERENCES 52
APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST 53
APPENDIX B – COURSE OUTCOME MAPPING 55
APPENDIX C – COMPLETE DELUGE SOURCE CODE 56
APPENDIX D – USER MANUAL 60
D.1 Accessing the Application 60
D.2 Instructions for Student / Faculty 60
D.3 Instructions for Administrator / Staff 60
APPENDIX E – EVALUATION RUBRIC 61
APPENDIX F – VIVA-VOCE RECORD SHEET 62

LIST OF TABLES

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application 4
Table 2.2: Summary of Literature Survey 5
Table 3.1: Stakeholders and User Roles 7
Table 3.2: Functional Requirements 7
Table 3.3: Non-Functional Requirements 8
Table 3.4: Hardware and Software Requirements 8
Table 3.5: Use Case Description – Request and Borrow a Reusable Item 9
Table 4.1: Entities and Relationships 11
Table 4.2: Field Specification – Campus_Users Form 12
Table 4.3: Field Specification – Item_Categories Form 13
Table 4.4: Field Specification – Reusable_Items Form 13
Table 4.5: Field Specification – Reuse_Requests Form 14
Table 4.6: Field Specification – Item_Loans Form 14
Table 4.7: Field Specification – Item_Returns Form 14
Table 4.8: Field Specification – Maintenance_Requests Form 15
Table 4.9: Field Specification – Item_Transfers Form 15
Table 4.10: Field Specification – Item_Donations Form 16
Table 4.11: Role–Permission Matrix 18
Table 5.1: Reports Implemented 29
Table 5.2: Deluge Scripts Implemented 31
Table 5.3: Workflows Implemented 36
Table 5.4: Integrations Implemented 39
Table 6.1: Test Cases 43
Table 6.2: Defect Log 44
Table 6.3: User Acceptance Feedback 44
Table 7.1: Deployment Checklist 45
Table 7.2: Performance Optimisation Applied 46
Table 7.3: Security Controls Implemented 46
Table 8.1: Mapping of Objectives to Outcomes 49
Table 8.2: Existing System versus Proposed Application 50
Table A.1: Feature Coverage Checklist 53
Table B.1: Mapping of Course Outcomes to Report Chapters 55
Table E.1: Evaluation Rubric for the Individual Project 61
Table E.2: Performance Level Descriptors 61
Table F.1: Viva-Voce Questions and Responses 62
Table F.2: Marks Summary 62

LIST OF FIGURES

Figure 2.1: As-Is Process Flow of the Existing System 4
Figure 3.1: Use Case Diagram of the Application 9
Figure 4.1: System Architecture of the Application 10
Figure 4.2: Entity–Relationship Diagram 11
Figure 4.3: Workflow / Process Flow for Reuse Request and Loan 17
Figure 4.4: Navigation Structure of the Application 18
Figure 5.1: Application Structure in the Design View (Forms, Reports, Workflows, Pages and Chats) 19
Figure 5.2: Campus_Users Form in Live Mode 20
Figure 5.3: Item_Categories Form 21
Figure 5.4: Reusable_Items Form – Item Details 22
Figure 5.5: Reuse_Requests Form – Request Details 23
Figure 5.6: Item_Loans Form – Loan Details 24
Figure 5.7: Item_Returns Form – Return Details 25
Figure 5.8: Maintenance_Requests Form – Issue Details 26
Figure 5.9: Item_Transfers Form – Transfer Details 27
Figure 5.10: Item_Donations Form – Donor and Item Details 28
Figure 5.11: Reports Listed in the Application Builder 29
Figure 5.12: Dashboard 31
Figure 5.13: Form Workflows in the Workflow Builder 37
Figure 5.14: Schedules 37
Figure 5.15: Approval 37
Figure 5.16: Blueprints List – Delivery Process 37
Figure 5.17: Delivery Process Blueprint Configuration 38
Figure 5.18: Batch Workflows List 38
Figure 5.19: Functions List 38
Figure 5.20: Zoho Invoice Connection Details 39
Figure 5.21: Connection Reference – Reusable_Items Form 40
Figure 5.22: Roles and Permission Profiles Configuration 40
Figure 5.23: Circular Campus Reuse Assistant (AI Agent) 41
Figure 5.24: Application on Mobile 42
Figure 7.1: Environments / Publish to Production 45
Figure 7.2: Audit Trail / Administration Settings 47
Figure 8.1: Live URL 49

LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
CRE / CRES	Campus Reusable Exchange (Network / System)
CRUD	Create, Read, Update, Delete
ER	Entity–Relationship
FR / NFR	Functional / Non-Functional Requirement
KPI	Key Performance Indicator
OAuth	Open Authorization
RBAC	Role-Based Access Control
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Educational campuses generate and consume a large number of physical resources every semester: laboratory coats, lab equipment, chargers, ergonomic chairs, stools, books, project components and office furniture. Many of these items are bought new for one academic term and then discarded, stored away or forgotten once the student or department no longer needs them. This is contrary to the principle of the circular economy, which aims to keep products and materials in use for as long as possible [5].

The Campus Reusable Exchange Network (CRES), also presented in the application dashboard as the "Circular Campus Smart Reusable Item Management System", is a Zoho Creator application that lets students, faculty and staff of a campus donate, list, request, borrow, return, repair and transfer reusable items through a single, governed platform. The application records every item, every person involved and every movement of an item, and rewards reuse through a Sustainability_Points measure stored against each campus user.

1.2 Problem Statement

Campus reuse is currently informal. Items are exchanged by word of mouth, messaging groups or paper registers, so nobody has a reliable view of what is available, who holds which item, when it is due back and whether it was returned damaged. A structured, role-based and automated system is needed to make reuse easy, accountable and measurable.

- Item availability is not visible; the same item is requested from several people and duplicate purchases happen.
- Requests, approvals and hand-overs have no audit trail; responsibility for loss or damage cannot be fixed.
- Due dates are tracked manually, so overdue items are noticed late and reminders are inconsistent.
- Damaged items are not routed for maintenance and are issued again in poor condition.
- There is no consolidated report of users, loans, donations and environmental benefit for campus management.

1.3 Objectives

- To design a normalised data model and forms for users, categories, reusable items, requests, loans, returns, maintenance, transfers and donations.
- To automate request validation, approval notification, status updates, overdue checks and stock alerts using Deluge scripting and workflows.
- To provide a role-aware dashboard and reports showing users, items, requests, loans and recent activity.
- To integrate the application with Zoho Invoice for item-related invoices and with Zia AI features (AI fields, object detection and an AI agent) for smart item handling.
- To publish the application securely with user, role and governance controls.

1.4 Scope of the Project

In scope: nine forms (Campus_Users, Item_Categories, Reusable_Items, Reuse_Requests, Item_Loans, Item_Returns, Maintenance_Requests, Item_Transfers, Item_Donations), their reports, one Dashboard page, six form workflows, two schedules, one approval process, one Blueprint, one batch workflow, one custom function, the Zoho Invoice connection and the "Circular Campus Reuse Assistant" AI agent. Users are students, faculty, staff and administrators of one campus, accessing the application on the web and the Zoho Creator mobile app.

Out of scope: online payment collection, barcode/RFID scanning, a public customer portal and pro-code extensions such as Zoho Catalyst functions.

1.5 Justification for a Low-Code Approach

Low-code platforms let business applications be assembled visually from forms, reports and workflows, and extended with a small amount of script where needed [1], [6], [7]. Zoho Creator suits CRES because the data model is form-centric, the processes (request → approval → loan → return) map directly onto workflows and Blueprints, hosting, security and mobile access are built in, and Zia AI and Zoho Invoice are available without custom integration code [2], [3]. The application was built by a single student in a few weeks, and can be maintained by non-programmers because most behaviour is configured rather than coded.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing arrangement a person who needs an item asks friends, posts in messaging groups or approaches a department directly. The owner agrees verbally, the transaction is noted (if at all) in a paper register or spreadsheet, and the item is handed over. The return date is remembered rather than tracked, and any damage is discovered only when the item is needed again, as shown in Figure 2.1.

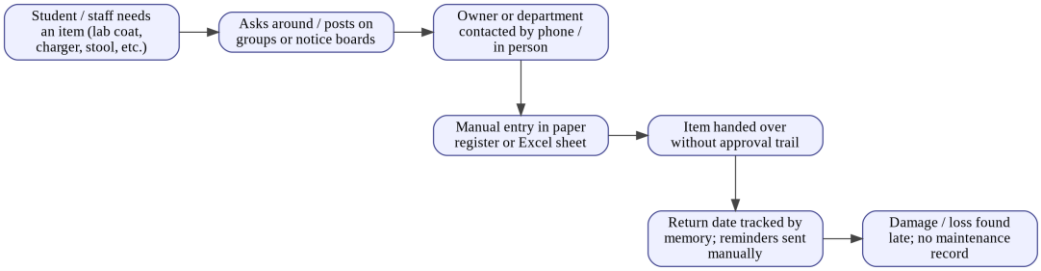


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- No single list of available items, quantities or conditions.
- No approval trail or accountability for hand-overs.
- No automatic due-date tracking, overdue detection or reminders.
- Damage and maintenance are not recorded, so poor-quality items are re-issued.
- No reports or dashboards; sustainability impact cannot be measured.
- Personal data and records are scattered across spreadsheets with no access control.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two approaches for CRES specifically.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Approx. 8–10 weeks (schema, API, UI, auth, deployment)	Approx. 2–3 weeks (forms, workflows, page configured visually)
Skills required	Full-stack developer, DBA, DevOps	Citizen developer with basic Deluge
Hosting & infrastructure	Servers, database and CI/CD to be set up	Fully managed by Zoho

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Mobile application	Separate native/hybrid app	Zoho Creator mobile app generated automatically
Security & access control	Custom authentication and authorisation code	Built-in users, roles and permissions
Maintenance & change	Code change, test and redeploy	Edit form/workflow and publish
Customisation limits	Unlimited	Bound by platform features and Deluge limits
Cost	Developer time plus hosting	Subscription (trial / education edition used)

2.4 Literature Survey

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	Ellen MacArthur Foundation, 2013 [5]	Towards the Circular Economy	Keeping products in use longer reduces waste and cost.	Motivation for campus item reuse.
2	Bock & Frank, 2021 [6]	Low-Code Platform	Defines low-code platforms and their model-driven, visual nature.	Basis for the low-code justification.
3	Sahay et al., 2020 [7]	Supporting the understanding and comparison of low-code development platforms	Compares features of low-code platforms.	Supports choice of Zoho Creator.
4	Waszkowski, 2019 [8]	Low-code platform for automating business processes in manufacturing	Shows workflow automation delivered quickly with low-code.	Similar approval and process automation.
5	Zoho Corporation [2], [3]	Zoho Creator Help; Deluge Reference	Documents forms, workflows, Blueprints, Zia and Deluge.	Implementation reference for all modules.

2.5 Emerging Trends in Low-Code Development

Current trends include AI-assisted application building, citizen development, hyper-automation and composable applications. CRES applies these ideas: Zia AI fields predict item category and keywords and verify images through object detection, an AI agent answers campus users in natural language, workflows and Blueprints automate the item life cycle, and Zoho Invoice is composed into the app as a ready-made service rather than being rebuilt.

2.6 Proposed System

The proposed CRES application replaces registers and chat groups with a single Zoho Creator application. A shared item catalogue removes the visibility problem; request, approval and loan forms provide the audit trail; schedules and workflows track due dates and send notifications; the maintenance and return forms capture condition so that damaged items are repaired before reuse; and the Dashboard provides live reporting. Each limitation in Section 2.2 is therefore addressed by a specific module described in Chapter 5.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Campus reuse administrators (5 on dashboard)	Manage users and categories, approve requests and donations, view all reports and dashboard
Faculty	Teaching staff (12 on dashboard)	Request and donate items, raise maintenance requests, approve where assigned
Staff	Non-teaching staff (8 on dashboard)	Issue items, record returns and inspections, handle maintenance and transfers
Student	Borrowers and donors (25 on dashboard)	Register, browse items, submit requests, donate, use the AI assistant

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall register campus users with name, college e-mail, role, department and register number.	Admin, Student	High
FR-02	The system shall maintain an item catalogue with category, condition, quantities and images.	Admin, Staff	High
FR-03	The system shall validate reuse requests before saving them.	All	High
FR-04	The system shall notify the approver by e-mail when a new request is created.	Approver	High
FR-05	The system shall record approval or rejection and update the request status automatically.	Admin	High
FR-06	The system shall issue items against approved requests and record due dates.	Staff	High
FR-07	The system shall record returns with condition, inspection notes and damage charge.	Staff	High
FR-08	The system shall record maintenance requests and e-mail the assigned approver; completing maintenance shall set the item available again.	Faculty, Staff	Medium
FR-09	The system shall record item transfers between campus locations.	Staff	Medium
FR-10	The system shall record donations and route them through an approval process.	Student, Admin	Medium
FR-11	The system shall flag overdue loans and send daily stock details by schedule.	System	Medium

FR ID	Requirement	Role(s)	Priority
FR-12	The system shall display a dashboard of KPIs, charts and latest updates.	Admin	Medium
FR-13	The system shall provide an AI assistant for finding items and raising requests.	All	Low
FR-14	The system shall connect to Zoho Invoice for item invoices.	Admin	Low

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Reports and dashboard shall load within a few seconds for the campus-scale data volume.
NFR-02	Security	Users shall access only the data and components permitted by their role.
NFR-03	Usability	The application shall be usable on the web and the Zoho Creator mobile app.
NFR-04	Availability	The application shall rely on Zoho's managed cloud hosting.
NFR-05	Maintainability	Changes shall be possible through configuration without redeploying code.
NFR-06	Auditability	Record additions and changes shall be traceable through the audit trail.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator Education Edition
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)
Other Zoho services	Zoho Invoice
Mobile	Zoho Creator mobile app (Android / iOS)
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

Figure 3.1 shows the use cases for each role. Table 3.5 describes the key use case of requesting and borrowing an item.

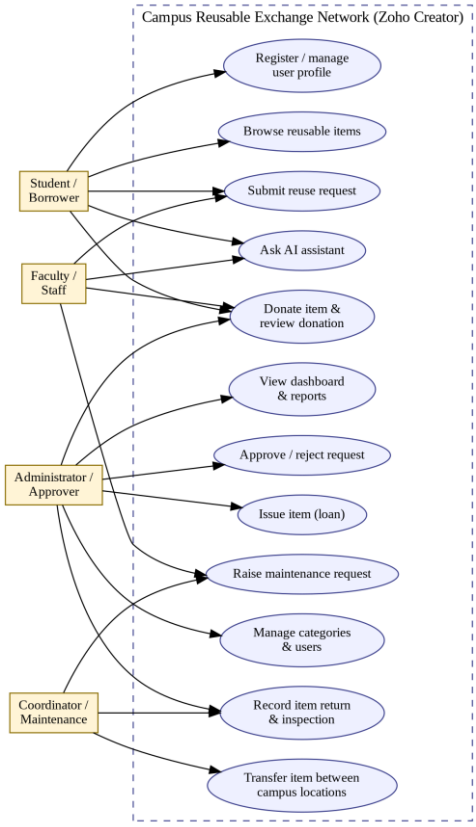


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Request and Borrow a Reusable Item

Field	Details
Use Case ID	UC-01
Use Case Name	Request and borrow a reusable item
Actor(s)	Student / Faculty, Administrator, Staff
Pre-condition	User is registered in Campus_Users and the item is available
Main Flow	1. User selects an item and submits the Reuse_Requests form. 2. Validate_Reuse_Request checks the data. 3. Approver is notified by e-mail. 4. Approver approves; Update_Approved_Status runs. 5. Staff issues the item through Item_Loans. 6. User returns the item; Item_Returns records condition.
Alternate Flow	Validation fails (alert shown) or approver rejects; Update_Rejected_Status runs and the request is closed.
Post-condition	Loan closed, stock updated, dashboard counts refreshed
Related FR	FR-03, FR-04, FR-05, FR-06, FR-07

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

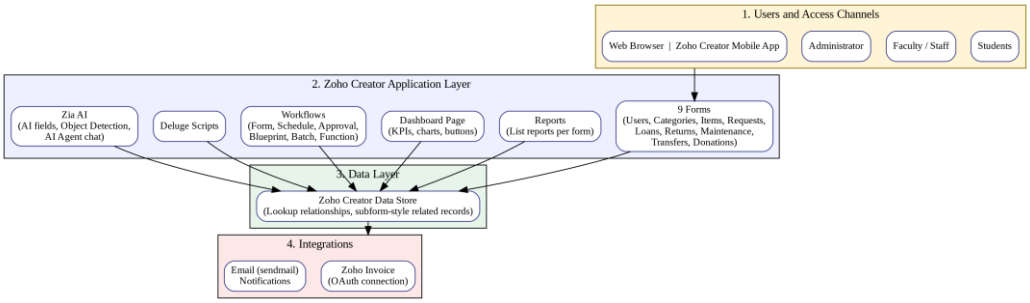


Figure 4.1: System Architecture of the Application

The architecture has four layers. Users (students, faculty, staff, administrators) reach the application through a web browser or the Zoho Creator mobile app. The application layer contains the forms, reports, the Dashboard page, workflows, Deluge scripts and Zia AI features. All of these read and write the Zoho Creator data store, where relationships between forms are realised through lookup fields and subform-style related records. The integration layer connects to Zoho Invoice through an authorised connection and sends notification e-mail through Deluge sendmail.

4.2 Data Model

Figure 4.2 shows the entity–relationship model. Item_Categories classify Reusable_Items; Campus_Users act as donors, requesters, borrowers, approvers and inspectors; a Reuse_Request produces an Item_Loan; each loan can have Item_Returns; items and loans may have Maintenance_Requests; and items may be moved by Item_Transfers or arrive through Item_Donations.

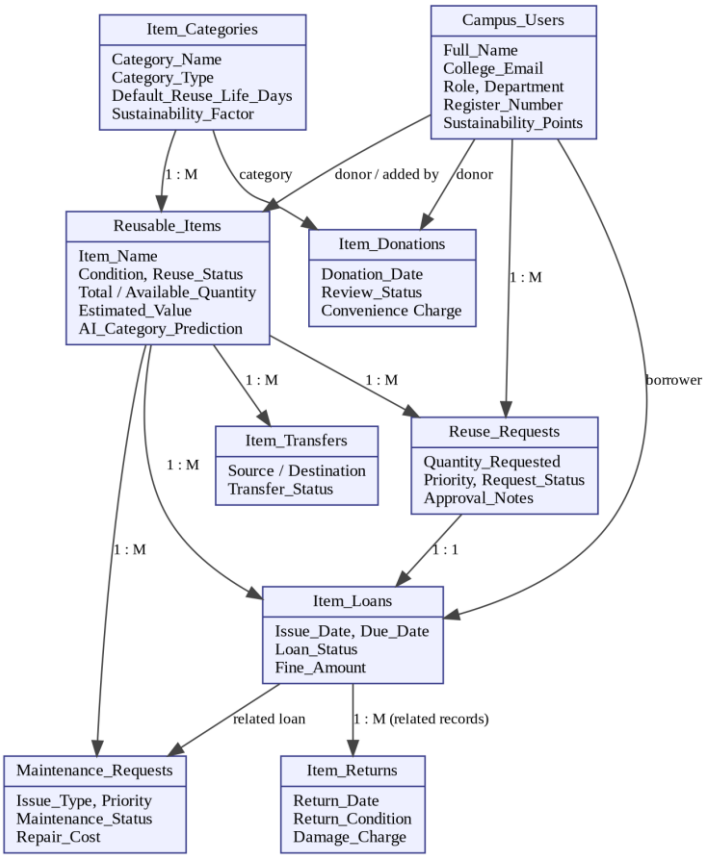


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Item_Categories	Category_Name, Category_Type, Sustainability_Factor	Reusable_Items	One-to-Many	Lookup (Category_Name) in Reusable_Items
Campus_Users	Full_Name, College_Email, Role, Register_Number	Reuse_Requests, Item_Loans, Item_Donations	One-to-Many	Lookups (Full_Name, Borrower, Donor_User)
Reusable_Items	Item_Name, Condition, Reuse_Status, Available_Quantity	Reuse_Requests, Item_Loans, Maintenance_Requests, Item_Transfers	One-to-Many	Lookups (Reusable_Items, Item)
Reuse_Requests	Quantity_Requested, Priority, Request_Status	Item_Loans	One-to-One	Lookup (Request) in Item_Loans
Item_Loans	Issue_Date, Due_Date, Loan_Status, Fine_Amount	Item_Returns, Maintenance_Requests	One-to-Many	Lookup (Loan, Related_Loan); related-record section “Item Return”

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Item_Returns	Return_Date, Return_Condition, Damage_Charge	Item_Loans	Many-to-One	Lookup (Loan)
Maintenance_Requests	Issue_Type, Priority, Maintenance_Status, Repair_Cost	Reusable_Items, Campus_Users	Many-to-One	Lookups (Reusable_Items, Assigned_To)
Item_Transfers	Quantity_Transferred, Transfer_Status	Reusable_Items, Campus_Users	Many-to-One	Lookups (Item, Requested_By)
Item_Donations	Donation_Date, Review_Status, Convenience Charge	Campus_Users, Item_Categories	Many-to-One	Lookups (Donor_User, Category)

4.3 Form Design

Field specifications for the main forms are given below, taken from the forms as built. Field types follow the Zoho Creator names. Only the fields visible in the form builder screens are listed; each form also holds system fields.

Table 4.2: Field Specification – Campus_Users Form

Field Label	Field Type	Mandatory	Validation / Remarks
Full_Name	Name	Yes	First and last name
User Name	Drop Down	No	Lookup-style selection
College_Email	Email	Yes	College e-mail used for notification
Phone_Number	Phone	No	Country code +91
Gender	Radio	No	Male / Female / Transgender
Role	Drop Down	No	Student / Faculty / Staff / Admin
Department	Drop Down	No	
Year_of_Study	Drop Down	Yes	
Register_Number	Single Line	Yes	
Campus_Location	Drop Down	No	
User_Status	Drop Down	No	
Profile_Photo	Image	No	
Joined_Date	Date	No	dd-MMM-yyyy
Total_Borrowed	Number	No	
Sustainability_Points	Number	No	Reuse reward points
Signature	Signature	No	Drawn on screen
How Do You Know About This?	Drop Down	No	

Table 4.3: Field Specification – Item_Categories Form

Field Label	Field Type	Mandatory	Validation / Remarks
Category_Name	Single Line	Yes	
Category_Description	Multi Line	No	
Category_Type	Drop Down	No	
Default_Reuse_Life_Days	Number	No	
Is_Active	Checkbox	Yes	Checked / Unchecked
Sustainability_Factor	Decimal	No	Used for impact scoring
ITEM MAINTENANCE	Subform (related records)	No	Reusable_Items, Related_Loan, Reported_By, Issue_Type

Table 4.4: Field Specification – Reusable_Items Form

Field Label	Field Type	Mandatory	Validation / Remarks
Item_Name	Single Line	Yes	
Category_Name	Lookup	No	Item_Categories
Borrower_Active_Loans	Number	Yes	
Item_Description	Multi Line	No	
Item_Type / Condition	Drop Down	Condition: Yes	
Total_Quantity / Available_Quantity	Number	Yes	Stock counts
Item_Location / Storage_Details	Single Line	No	
Ownership_Type	Drop Down	No	
Donor_User / Added_By	Lookup	No	Campus_Users
Date_Acquired	Date	No	
Total Duration Days	Number	No	
Estimated_Value	Currency	Yes	
Item_Image / Item_Document / About Item	Image / File Upload	No	About Item accepts file or video
Reuse_Status	Drop Down	No	Available / Maintenance, etc.
Condition_Notes	Multi Line	No	
AI_Category_Prediction / AI_Keywords	AI field (read-only)	No	prediction / keyword extraction
Object Detection Verification	AI field (read-only)	No	object detection
Last_Inspection_Date	Date	No	
Invoices	Integration field	No	Zoho Invoice

Table 4.5: Field Specification – Reuse_Requests Form

Field Label	Field Type	Mandatory	Validation / Remarks
Full_Name	Lookup	No	Campus_Users
Reusable_Items	Lookup	No	Reusable_Items
Quantity_Requested	Number	No	
Request_Date / Required_From / Expected_Return_Date	Date	Yes	
Purpose	Multi Line	No	
Priority	Drop Down	No	
Request_Status	Drop Down	No	Pending / Approved / Rejected / Cancelled
Approver	Lookup	No	Campus_Users
Approval_Notes	Multi Line	No	
Created_At	Date-Time	No	dd-MMM-yyyy HH:mm:ss

Table 4.6: Field Specification – Item_Loans Form

Field Label	Field Type	Mandatory	Validation / Remarks
Request / Borrower / Item	Lookup	No	Reuse_Requests, Campus_Users, Reusable_Items
Quantity_Issued	Number	No	
Issue_Date / Due_Date	Date	Yes	
Issued_By	Lookup	No	
Loan_Status	Drop Down	No	Active / Returned / Overdue / Lost
Actual_Return_Date	Date	No	
Extension_Count	Number	No	
Fine_Amount	Currency	No	
Loan_Notes	Multi Line	No	
ITEM RETURN	Subform (related records)	No	Loan, Returned_By, Return_Date, Quantity_Returned

Table 4.7: Field Specification – Item_Returns Form

Field Label	Field Type	Mandatory	Validation / Remarks
Loan / Returned_By	Lookup	No	
Return_Date	Date	No	
Quantity_Returned	Number	No	
Return_Condition	Drop Down	Yes	
Inspection_Notes	Multi Line	No	
Inspection_Photo	Image	No	

Field Label	Field Type	Mandatory	Validation / Remarks
Damage_Charge	Currency	No	
Inspected_By	Lookup	No	
Return_Status	Drop Down	No	
Stock_Updated	Checkbox	No	

Table 4.8: Field Specification – Maintenance_Requests Form

Field Label	Field Type	Mandatory	Validation / Remarks
Reusable_Items / Related_Loan / Reported_By	Lookup	No	
Issue_Type	Drop Down	No	
Problem_Description	Multi Line	No	
Problem_Photo	Image	No	
Priority	Drop Down	No	
Assigned_To	Lookup	No	Approver for e-mail
Report_Date / Target_Completion_Date	Date	No	
Maintenance_Status	Drop Down	No	Assigned / In Progress / Completed
Repair_Cost	Currency	No	
Resolution_Notes	Multi Line	No	
Maintenance_Recommendation	AI field (read-only)	No	Ai Field
Maintanance Steps	Drop Down	No	

Table 4.9: Field Specification – Item_Transfers Form

Field Label	Field Type	Mandatory	Validation / Remarks
Item	Lookup	No	
Quantity_Transferred	Number	No	
Source_Location / Destination_Location	Single Line	No	
Requested_By / Approved_By / Received_By	Lookup	No	
Transfer_Date / Received_Date	Date	No	
Transfer_Status	Drop Down	No	
Transfer_Notes	Multi Line	No	

Table 4.10: Field Specification – Item_Donations Form

Field Label	Field Type	Mandatory	Validation / Remarks
Donor_User / Donor_Name	Lookup	No	
Donor_Email	Email	No	
Item_Name / Category	Lookup	No	
Description	Multi Line	No	
Quantity	Number	No	
Condition	Checkbox	No	
Donation_Photo	Image	No	
Estimated_Value	Currency	No	
Convenience Charge	Formula (read-only)	No	Calculated
Donation_Date	Date	No	
Collection_Location	Single Line	No	
Review_Status	Checkbox	No	Completed / Not Completed
Reviewed_By	Lookup	No	
Review_Notes	Multi Line	No	
Review_Status Prediction	AI field (read-only)	No	AI Field

4.4 Process and Workflow Design

Figure 4.3 shows the end-to-end request process and where workflows run: validation on form submission, e-mail notification, approval, status update, loan, return and the scheduled overdue check.

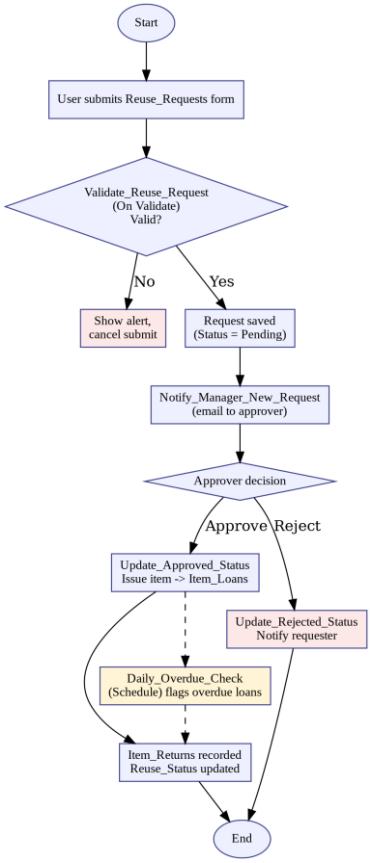


Figure 4.3: Workflow / Process Flow for Reuse Request and Loan

4.5 User Interface and Navigation Design

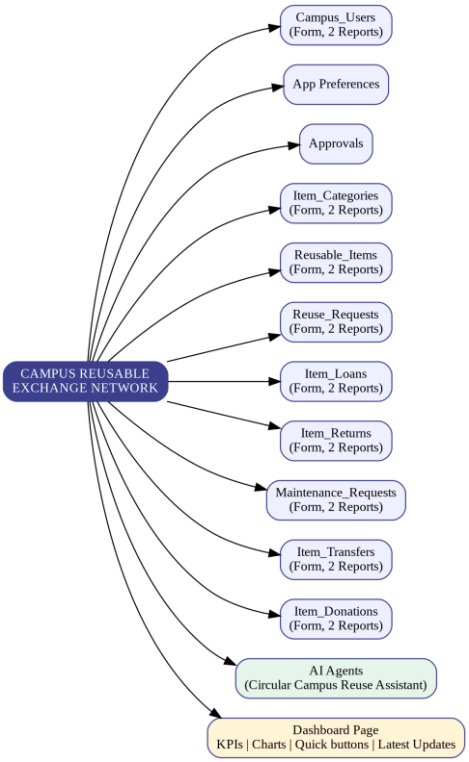


Figure 4.4: Navigation Structure of the Application

4.6 Access Control Design

Table 4.11: Role–Permission Matrix

Component	Administrator	Faculty	Staff	Student
Form: Campus_Users	A E V D	V	V	A V (own)
Form: Reusable_Items	A E V D	A V	A E V	V
Form: Reuse_Requests	A E V D	A V	E V	A V (own)
Form: Item_Loans / Item_Returns	A E V D	V	A E V	V (own)
Form: Maintenance_Requests	A E V D	A V	A E V	—
Form: Item_Donations	A E V D	A V	V	A V (own)
Page: Dashboard	V	V	V	—

A = Add, E = Edit, V = View, D = Delete, — = No access. This is the designed matrix; Section 5.8 records how it is configured.

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application “Campus Reusable Exchange Network” (short name CRE) was created from scratch in a Zoho Creator workspace on a trial edition. The left navigation lists a section for each form with its reports, plus App Preferences, Approvals, AI Agents and Dashboard, with a dark indigo theme and the CRE logo. Figure 5.1 shows the application builder with all forms, reports, workflows, pages and the AI chat.

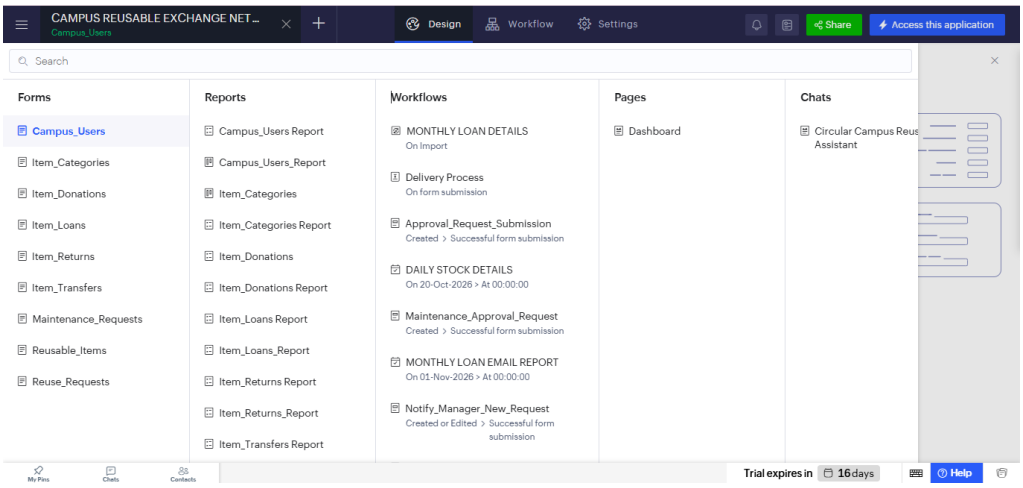


Figure 5.1: Application Structure in the Design View (Forms, Reports, Workflows, Pages and Chats)

5.2 Forms and Data Capture

Nine forms capture the data of the application. Each uses appropriate field types, lookups to related forms, mandatory markers (red asterisk) and a Submit button. The form screens are presented below form by form.

5.2.1 Campus_Users

Campus_Users stores the profile of every person: name, e-mail, phone, gender, role, department, year of study, register number, campus location, status, photo, joined date, borrowing count, sustainability points and a drawn signature.

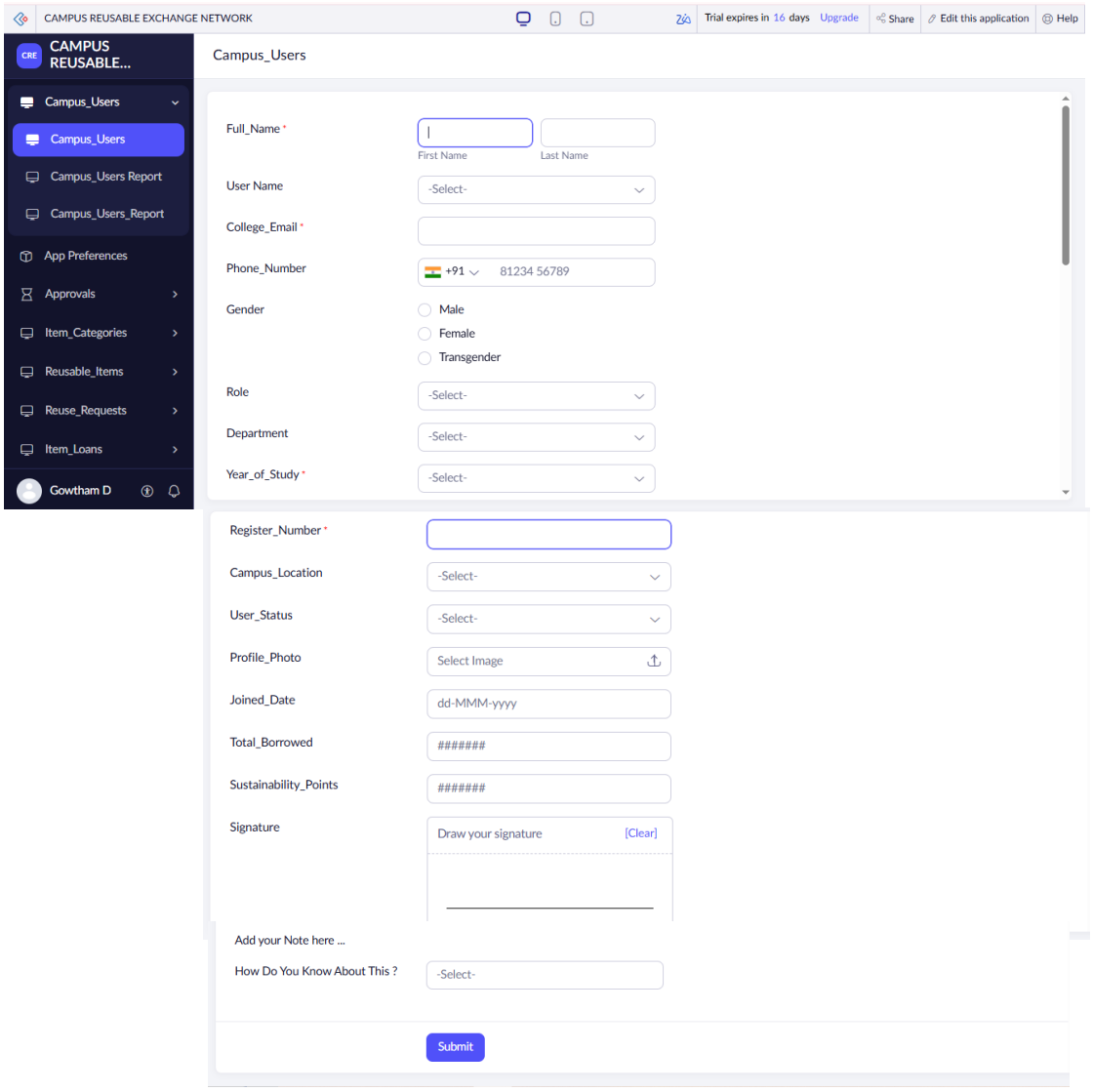


Figure 5.2: Campus_Users Form in Live Mode

5.2.2 Item_Categories

Item_Categories defines each category with a type, default reuse life in days, an Is_Active flag and a Sustainability_Factor. A subform-style “Item Maintenance” section lists related maintenance records.

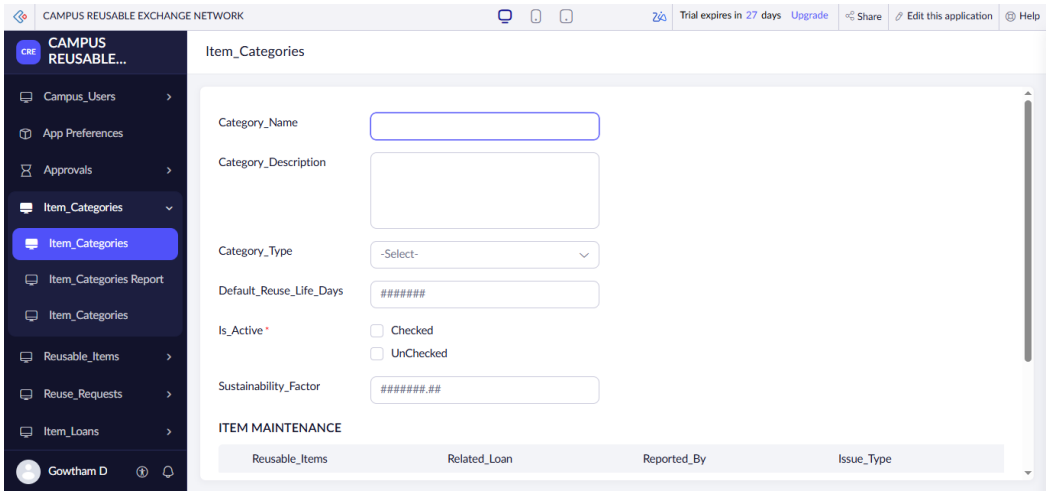
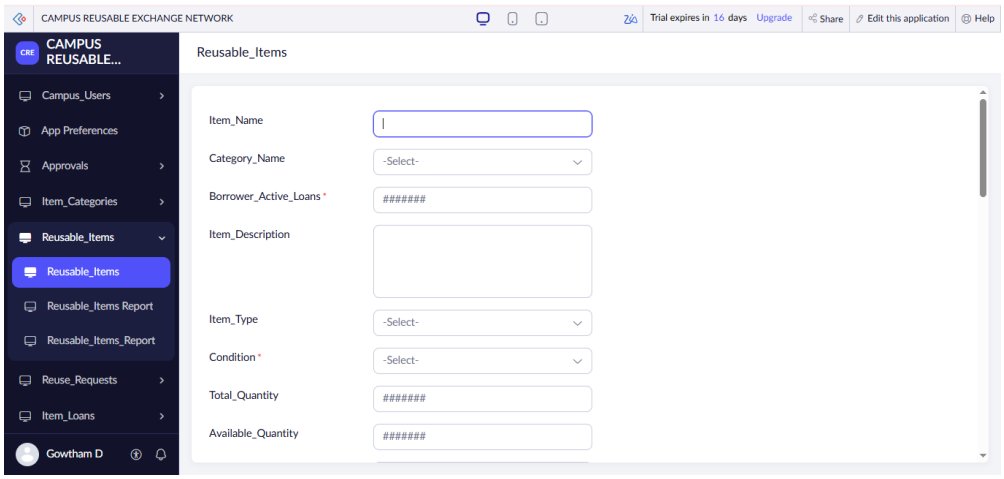


Figure 5.3: Item_Categories Form

5.2.3 Reusable_Items

Reusable_Items is the central catalogue. It uses a lookup to Item_Categories and Campus_Users, image and document uploads, currency and date fields, and read-only Zia AI fields (AI_Category_Prediction, AI_Keywords and Object Detection Verification). An integration field links to Zoho Invoice.



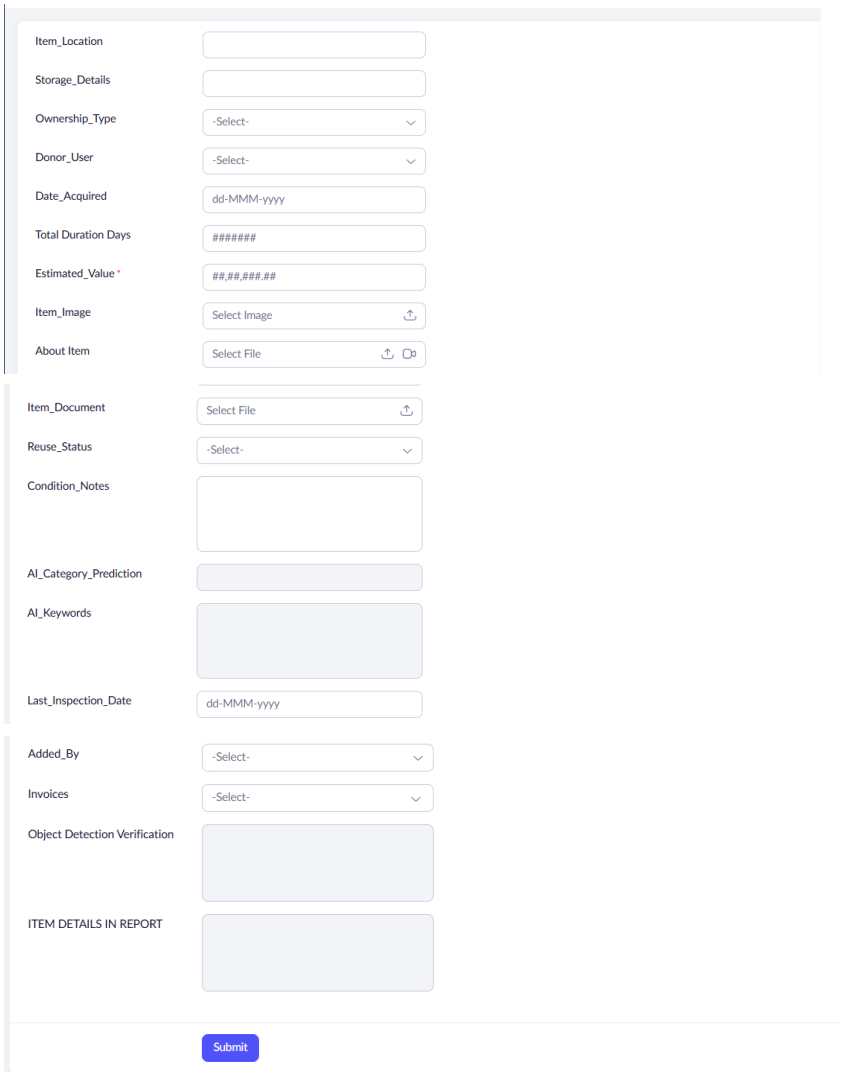


Figure 5.4: Reusable_Items Form – Item Details

5.2.4 Reuse_Requests

Reuse_Requests records who wants which item, quantity, dates, purpose, priority, status, approver and approval notes. Its workflows validate the data and notify the approver.

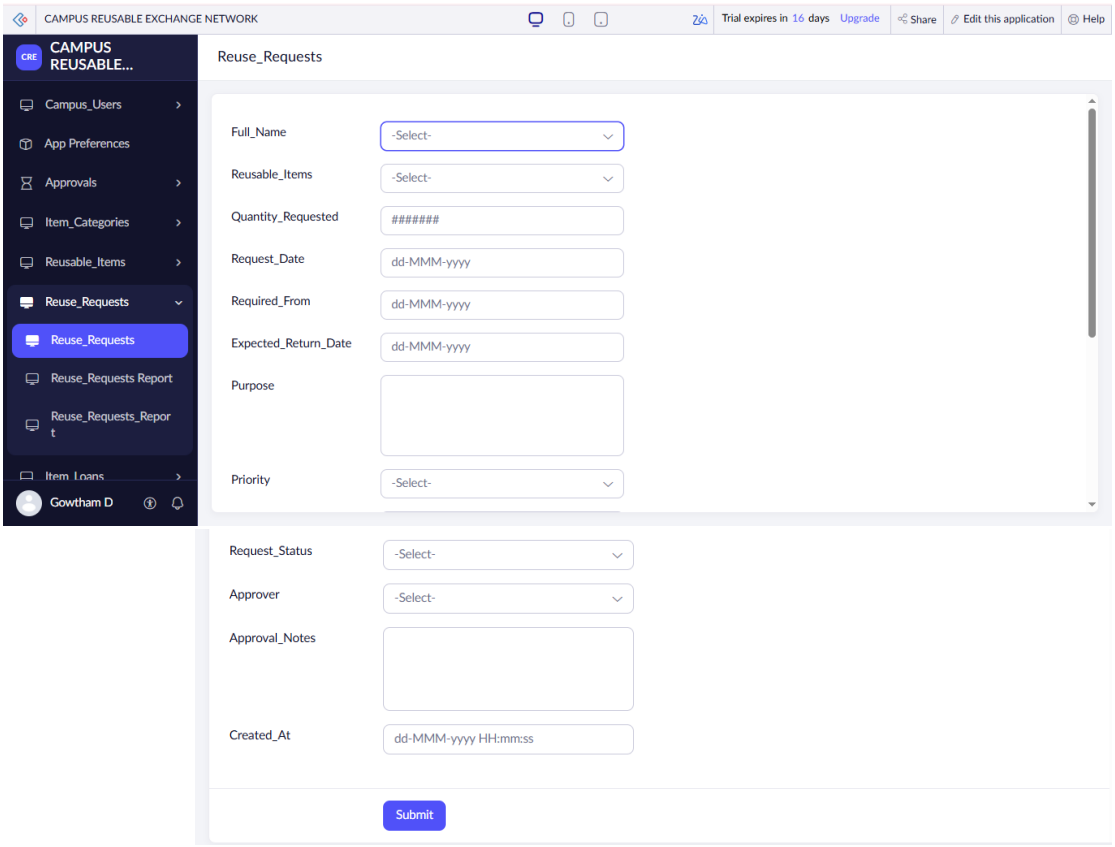


Figure 5.5: Reuse_Requests Form – Request Details

5.2.5 Item_Loans

Item_Loans converts an approved request into a loan with issue date, due date (both mandatory), status, extension count and fine amount. A related “Item Return” section lets returns be added directly from the loan.

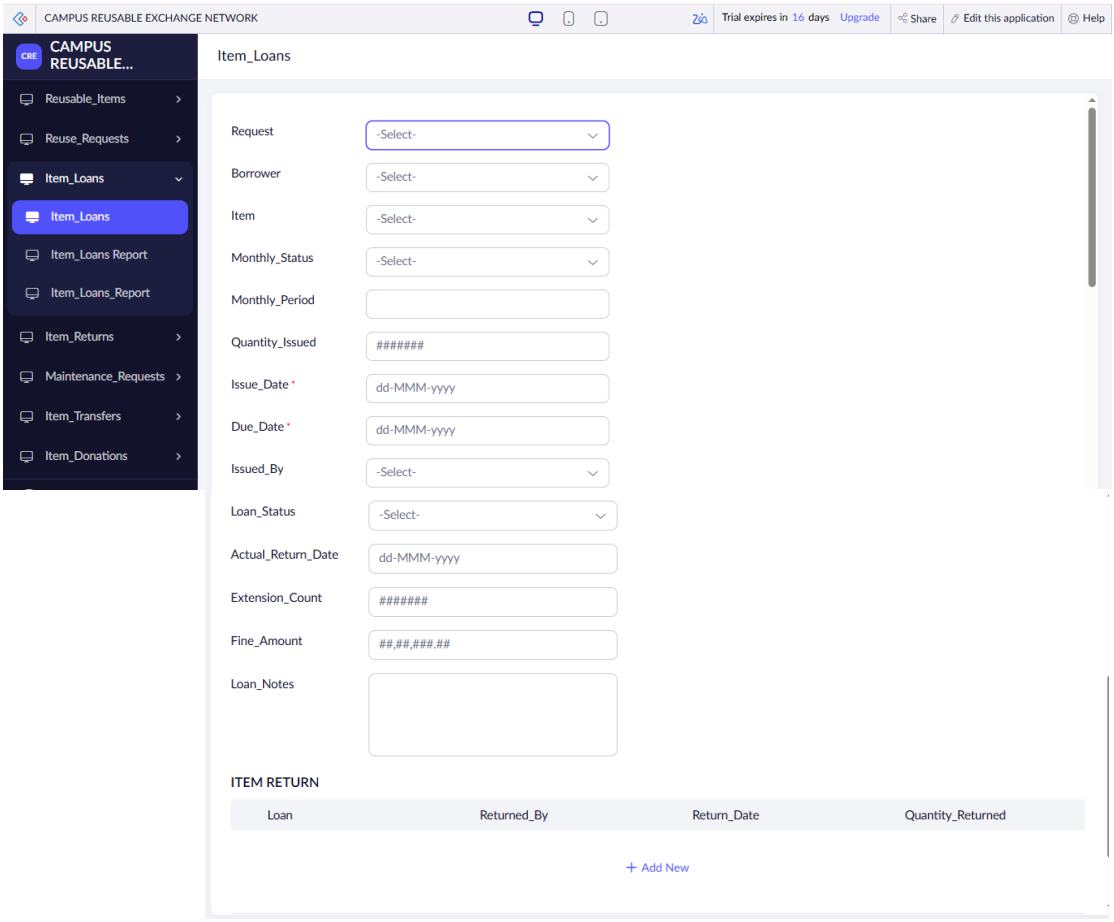


Figure 5.6: Item_Loans Form – Loan Details

5.2.6 Item_Returns

Item_Returns captures the return with condition, inspection notes, inspection photo, damage charge, inspector and a Stock_Updated flag.

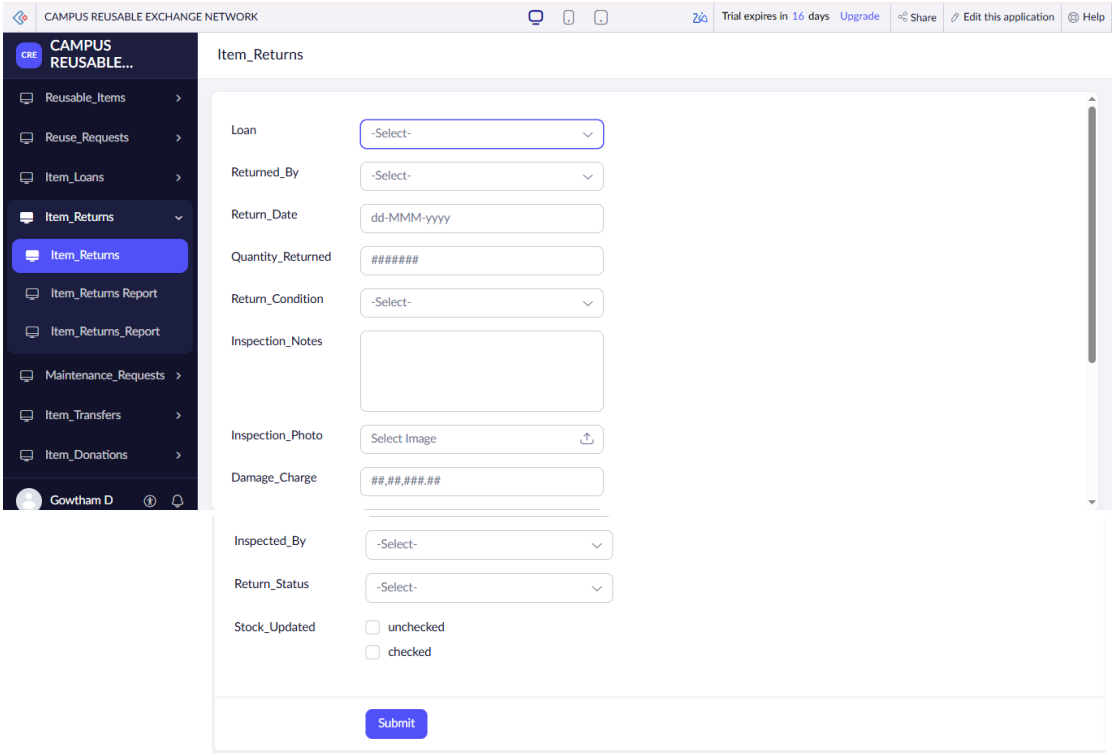


Figure 5.7: Item_Returns Form – Return Details

5.2.7 Maintenance_Requests

Maintenance_Requests logs issues with an item, with problem description and photo, priority and an assigned person, followed by status, repair cost, resolution notes and an AI-generated Maintenance_Recommendation.

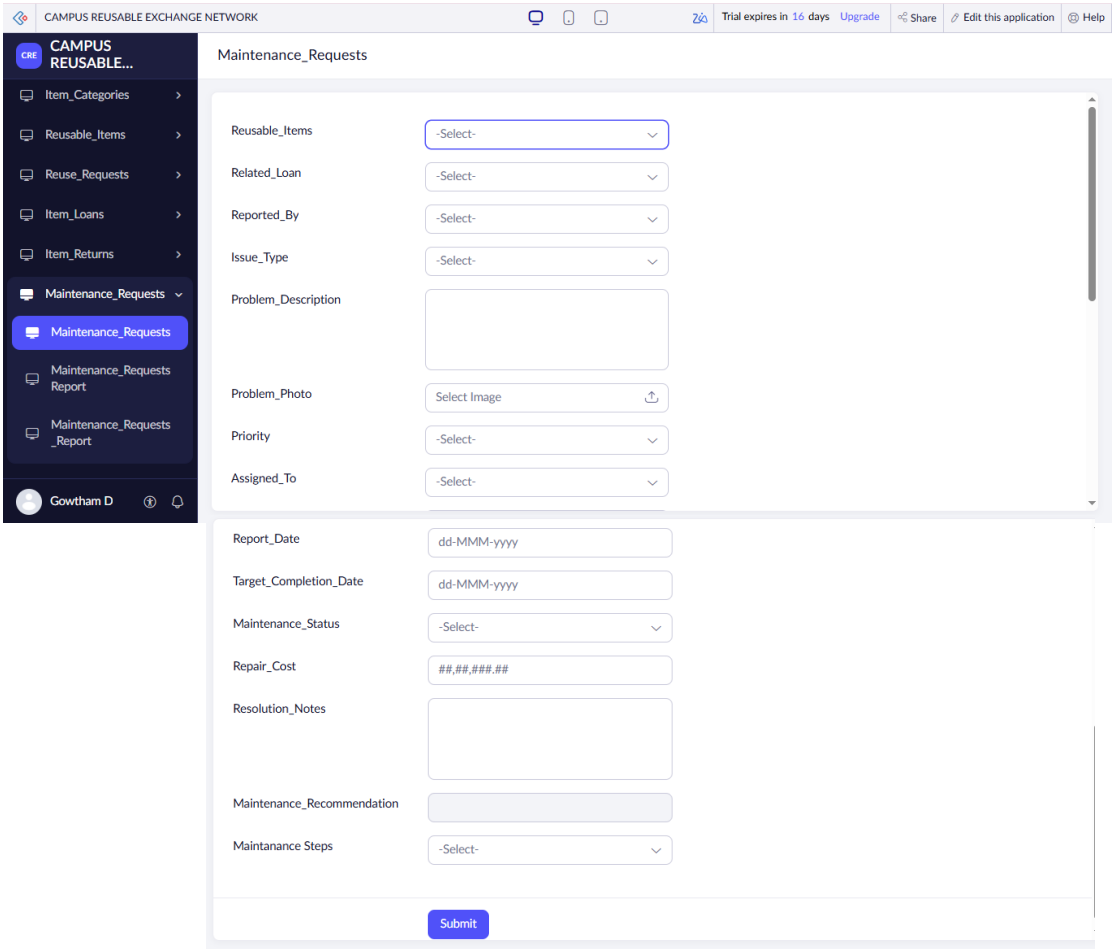


Figure 5.8: Maintenance_Requests Form – Issue Details

5.2.8 Item_Transfers

Item_Transfers moves stock between campus locations with source, destination, requester, approver, receiver, dates and status.

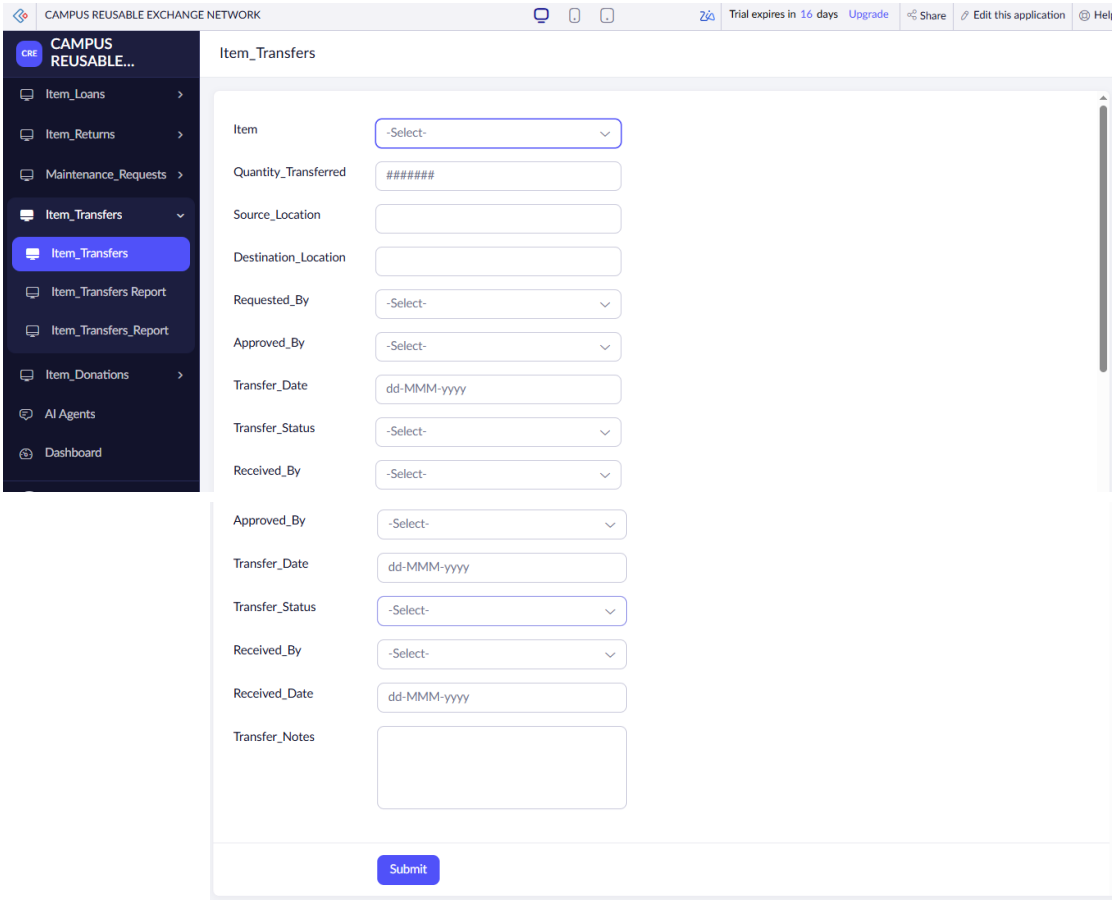


Figure 5.9: Item_Transfers Form – Transfer Details

5.2.9 Item_Donations

Item_Donations lets a donor offer an item with photo, estimated value and a calculated convenience charge. A reviewer completes the review, and a Zia field predicts the review status.

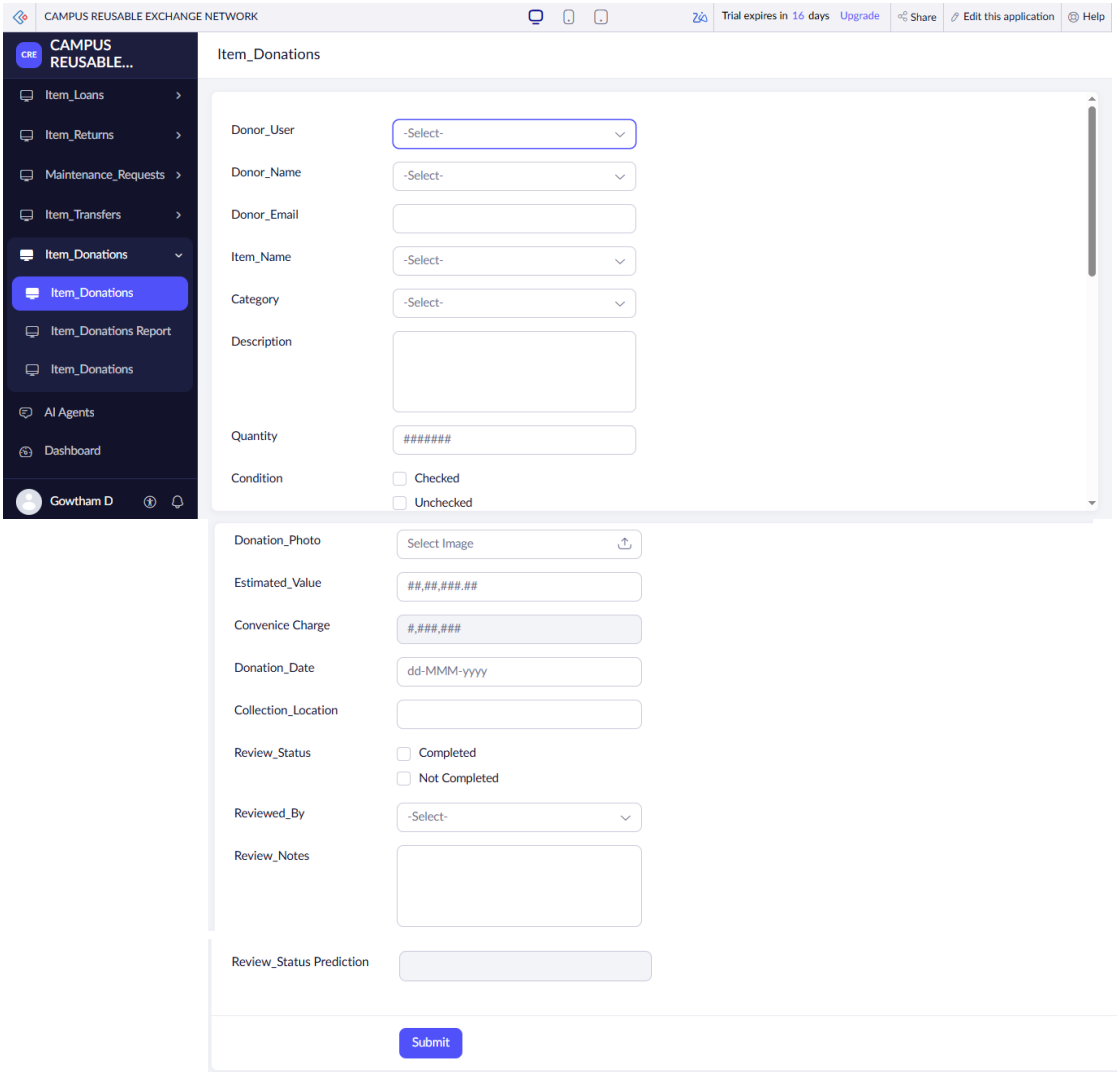


Figure 5.10: Item_Donations Form – Donor and Item Details

5.3 Reports

Every form has two list reports (for example Campus_Users Report and Campus_Users_Report), giving 18 reports in total, visible in the application structure (Figure 5.11). They present the records of each form in tabular form with quick view and detail view.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Roles
Campus_Users Report	List	Campus_Users	Admin
Item_Categories Report	List	Item_Categories	Admin
Reusable_Items_Report	List	Reusable_Items	All
Reuse_Requests_Report	List	Reuse_Requests	Admin, requester
Item_Loans_Report	List	Item_Loans	Admin, Staff
Item_Returns_Report	List	Item_Returns	Admin, Staff
Maintenance_Requests_Report	List	Maintenance_Requests	Admin, Staff
Item_Transfers_Report	List	Item_Transfers	Admin, Staff
Item_Donations Report	List	Item_Donations	Admin

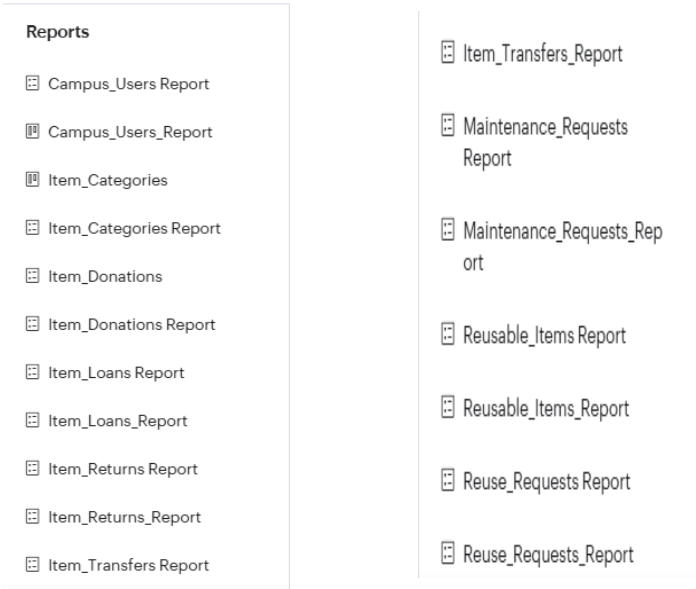
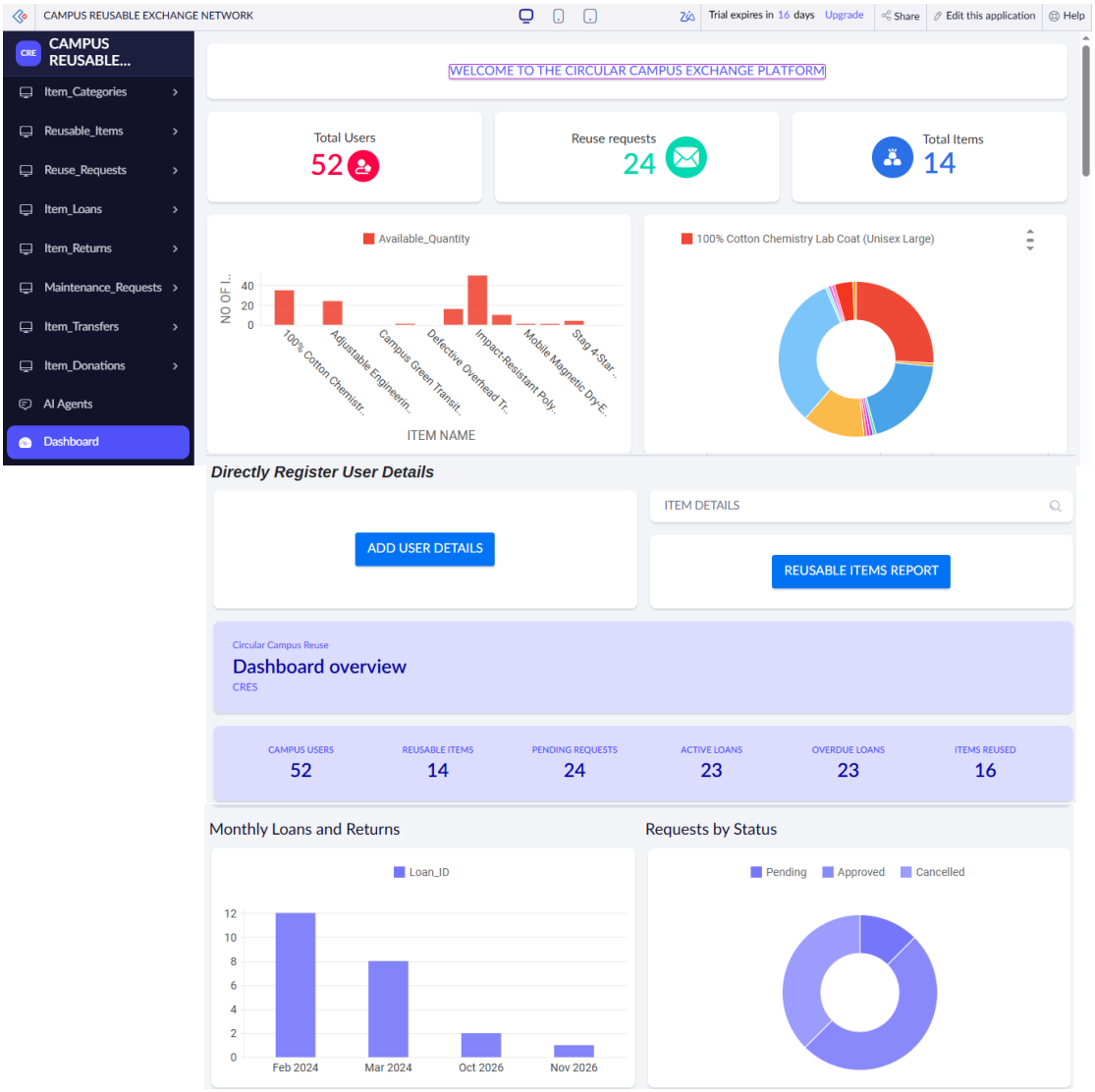


Figure 5.11: Reports Listed in the Application Builder

5.4 Pages and Dashboards

The Dashboard page, built in the UI Builder, is the landing screen for managers. The top shows a welcome banner and three KPI panels (Total Users 50, Reuse requests 15, Total Items 13) followed by a bar chart of available quantity by item and a donut chart. Quick buttons “Add User Details” and “Reusable Items Report” and an item search box allow direct actions. A KPI strip shows Campus Users 50, Reusable Items 13, Pending Requests 15, Active Loans 20, Overdue Loans 20 and Items Reused 15. Charts show Monthly Loans and Returns, Requests by Status, Users by Role, Items by Category, an Overdue Loans panel and a Latest Updates feed.



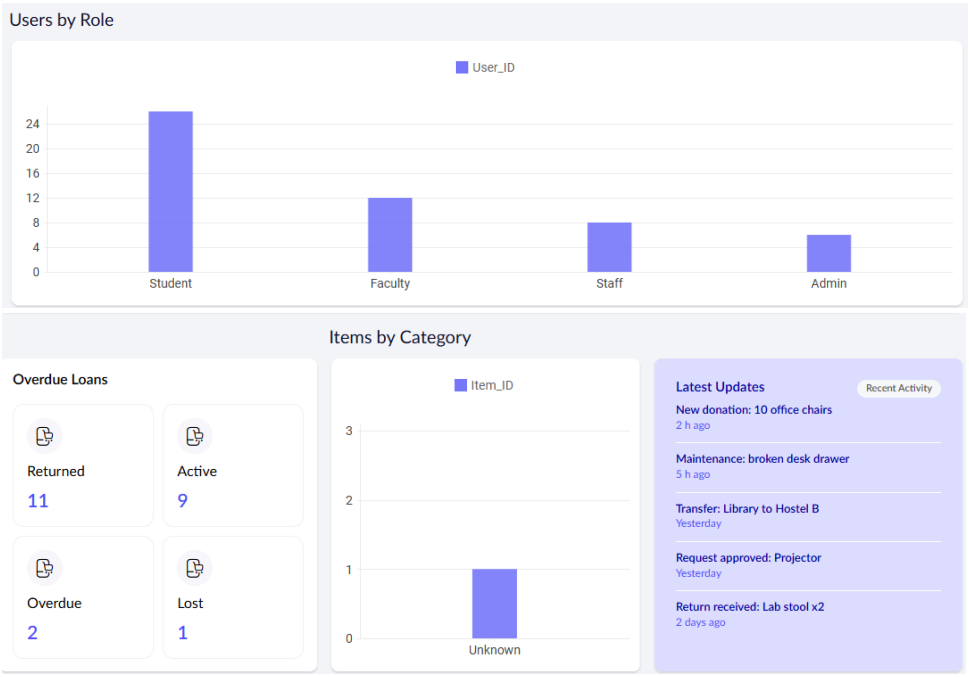


Figure 5.12: Dashboard

5.5 Deluge Scripting

Deluge scripts automate notifications, status changes, cancellation and overdue checks. Table 5.2 summarises them; the full code is in Appendix C.

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Maintenance_Requests → Created, Successful form submission (Maintenance_Approval_Request)	E-mail the assigned approver about a new maintenance request	fetch by criteria, null checks, string concatenation, sendmail, info
DS-02	Maintenance_Requests → Edited, Update of Maintenance_Status (Maintenance_Completion_Update)	Set item status Available on completion, Maintenance when work begins	if–else if, fetch, record update
DS-03	Function: cancelOrder(int requestID, string reason)	Cancel a request unless already cancelled, approved or completed	fetch, count, nested if, return, string update
DS-04	Schedule: Daily_Overdue_Check	Find active loans past their due date	zoho.currentdate, for each with criteria

Code 5.1: Maintenance approval e-mail (DS-01)

```
sendmail
[
	from :zoho.adminuserid
	to :"gowthamdevendran1406@gmail.com"
	subject :"New Maintenance Approval Request - Circular Campus"
	message :"A new maintenance request has been submitted in Circular
Campus.\n\n" + "Maintenance ID: " + input.Maintenance_ID + "\n" + "Issue Type:
" + input.Issue_Type + "\n" + "Priority: " + input.Priority + "\n" + "Status: " +
input.Maintenance_Status + "\n\n" + "Please log in to Circular Campus and review
the maintenance request."
]
```

DS-01 fetches the approver from the Assigned_To lookup, checks that a college e-mail exists, builds an HTML message with the maintenance ID, item, issue type, priority and description, and sends it. If no approver or e-mail exists it logs the reason with info instead of failing.

Code 5.2: cancelOrder custom function (DS-02)

```
void cancelOrder(int requestID, string reason)
{
	// Find request using Request_ID
	req = Reuse_Requests[Request_ID == requestID];
	if(req.ID == null)
	{
		info "Request not found.";
		return;
	}
	// Check request status
	if(req.Request_Status == "Cancelled")
	{
		info "Request is already cancelled.";
		return;
	}
	if(req.Request_Status == "Approved" || req.Request_Status ==
"Completed")
	{
		info "This request cannot be cancelled.";
		return;
	}
	// Find the linked Campus_Users record
	userRecord = Campus_Users[ID == req.Full_Name];
	if(userRecord.ID == null)
	{
		info "Linked Campus User not found.";
		return;
	}
	// Get registered email
```

```
	userEmail = userRecord.College_Email;
	if(userEmail == null || userEmail == "")
	{
		info "User email address is missing.";
		return;
	}
	// Set cancellation reason
	if(reason == null || reason.trim() == "")
	{
		reason = "No reason provided";
	}
	// Update request status
	req.Request_Status="Cancelled";
	req.Approval_Notes="Cancelled: " + reason;
	// Send email to the selected user
	sendmail
	[
		from :zoho.adminuserid
		to :userEmail
		subject :"Circular Campus - Reuse Request Cancelled"
		message :"Hello " + userRecord.Full_Name + ",\n\nYour reuse
request has been cancelled successfully.\n\nRequest ID: " + req.Request_ID +
"\nQuantity: " + req.Quantity_Requested + "\nReason: " + reason + "\nStatus:
Cancelled\n\nThank you,\nCircular Campus Team"
	]
	info "Request cancelled successfully.";
	info "Email sent to: " + userEmail;
}
```

DS-02 guards against invalid cancellation: it reports if the request is missing, already cancelled, or already approved or completed; otherwise it sets Request_Status to Cancelled and appends the reason to Approval_Notes.

Code 5.3: MONTHLY LOAN DETAILS (DS-03)

```
loanCount = 1;
returnedCount = 0;
pendingCount = 0;
overdueCount = 0;
fineAmount = 0.0;
if(input.Loan_Status == "Returned")
{
	returnedCount = 1;
}
else if(input.Loan_Status == "Pending")
{
	pendingCount = 1;
}
```

```
else if(input.Loan_Status == "Overdue")
{
	overdueCount = 1;
}
if(input.Fine_Amount != null)
{
	fineAmount = input.Fine_Amount;
}
info "Monthly Loan Record Processed";
info "Loan ID: " + input.ID;
info "Total Loans: " + loanCount;
info "Returned: " + returnedCount;
info "Pending: " + pendingCount;
info "Overdue: " + overdueCount;
info "Fine: " + fineAmount;
```

Code 5.4 : Reuse Request Approval (DS-04)

For Approval :

```
requestRecord = Reuse_Requests[ID == input.ID];
if(requestRecord != null)
{
	requestRecord.Request_Status="Approved";
	requestRecord.Approver=zoho.loginuser;
	requestRecord.Approval_Notes="Request approved by " + zoho.loginuser;
	info "Request approved successfully.";
}
```

For Reject :

```
requestRecord = Reuse_Requests[ID == input.ID];
if(requestRecord != null)
{
	requestRecord.Request_Status="Approved";
	requestRecord.Approver=zoho.loginuser;
	requestRecord.Approval_Notes="Request approved by " + zoho.loginuser;
	info "Request approved successfully.";
}
```

Code 5.5 : Monthly Loan Details Report (DS-05)

```
previousMonthDate = zoho.currentdate.toStartOfMonth().addMonth(-1);
reportMonth = previousMonthDate.toString("MMMM-yyyy");
reportMonthNumber = previousMonthDate.getMonth();
reportYear = previousMonthDate.getYear();
totalLoans = 0;
returnedLoans = 0;
pendingLoans = 0;
overdueLoans = 0;
totalFines = 0.0;
for each  loan in Item_Loans
{
	if(loan.Issue_Date != null)
	{
		if(loan.Issue_Date.getMonth() == reportMonthNumber &&
loan.Issue_Date.getYear() == reportYear)
		{
			totalLoans = totalLoans + 1;
			if(loan.Loan_Status == "Returned")
			{
				returnedLoans = returnedLoans + 1;
			}
			else if(loan.Loan_Status == "Pending")
			{
				pendingLoans = pendingLoans + 1;
			}
			else if(loan.Loan_Status == "Overdue")
			{
				overdueLoans = overdueLoans + 1;
			}
			if(loan.Fine_Amount != null)
			{
				totalFines = totalFines + loan.Fine_Amount;
			}
		}
	}
}
emailMessage = "CIRCULAR CAMPUS - MONTHLY LOAN REPORT\n\n";
emailMessage = emailMessage + "Report Month: " + reportMonth + "\n\n";
emailMessage = emailMessage + "TOTAL LOAN DETAILS\n";
emailMessage = emailMessage + "Total Loans: " + totalLoans + "\n";
emailMessage = emailMessage + "Returned Loans: " + returnedLoans + "\n";
emailMessage = emailMessage + "Pending Loans: " + pendingLoans + "\n";
emailMessage = emailMessage + "Overdue Loans: " + overdueLoans + "\n";
emailMessage = emailMessage + "Total Fine Amount: Rs. " + totalFines + "\n\n";
emailMessage = emailMessage + "This is the automatically generated monthly
report from Circular Campus.";
sendmail
[
```

```
        from :zoho.adminuserid
        to :"gowtham.d.2024.cse@rajalakshmi.edu.in"
        subject :"Circular Campus - Monthly Loan Report - " + reportMonth
        message :emailMessage
]
info "Monthly loan report email sent.";
info emailMessage;
```

5.6 Workflows and Automation

Table 5.3 lists every workflow. Form workflows react to submissions, schedules run by date, the approval process governs donations, the Blueprint governs delivery, and the batch workflow processes imported loans.

Table 5.3: Workflows Implemented

Workflow Name	Type	Form	Trigger / Event	Action(s)	Script Ref.
Maintenance_Approval_Request	Form	Maintenance_Requests	Created > Successful form submission	Mail approver	DS-01
DAILY STOCK DETAILS	Schedule	—	20-Oct-2026, 00:00	Send stock mail	—
Montly Loan Details Report	Schedule	—	30-Oct-2026,00:00	Monthly Loan Report	DS - 05
Reuse Request Approval	Approval	Reuse Request	On review	Mail submitter on approve / reject	DS - 04
Delivery Process	Blueprint	—	Stage transitions	Route item delivery	—
MONTHLY LOAN DETAILS	Batch	Item_Loans	After successful import, 5 records per batch	Per-record action	DS-03
cancelOrder	Function	Reuse_Requests	Called from script	Cancel request	DS-02

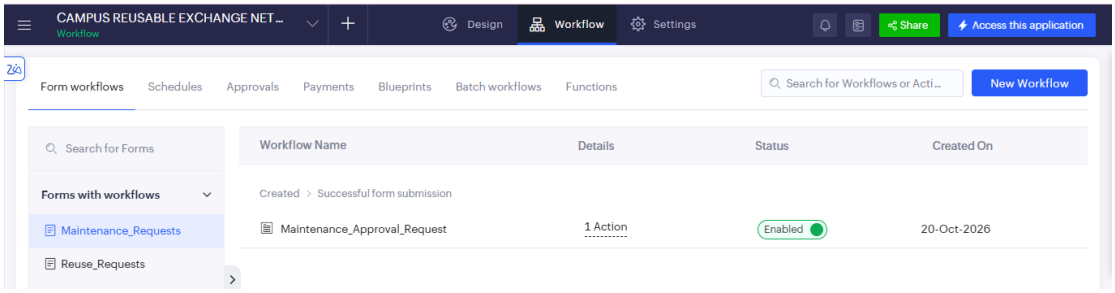


Figure 5.13: Form Workflows in the Workflow Builder

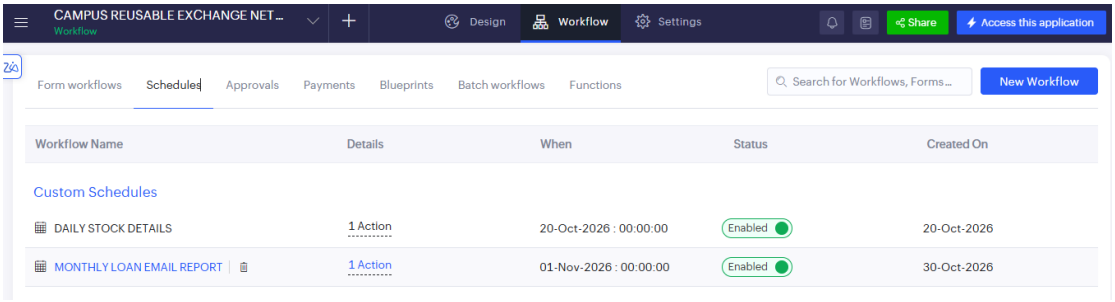


Figure 5.14: Schedules

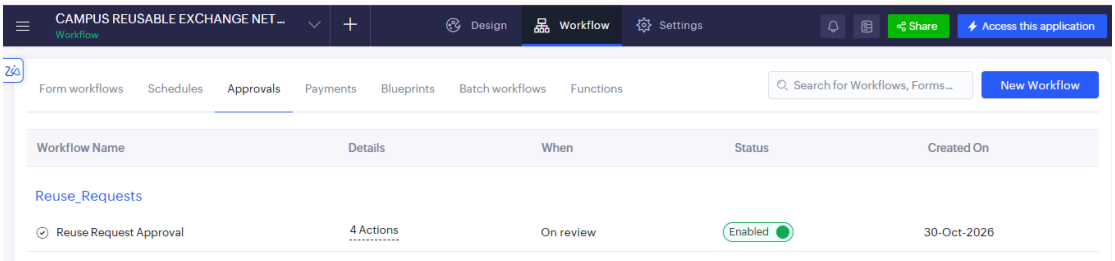


Figure 5.15: Approval

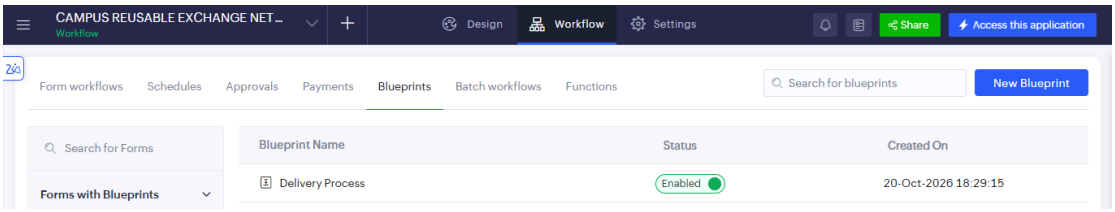


Figure 5.16: Blueprints List – Delivery Process

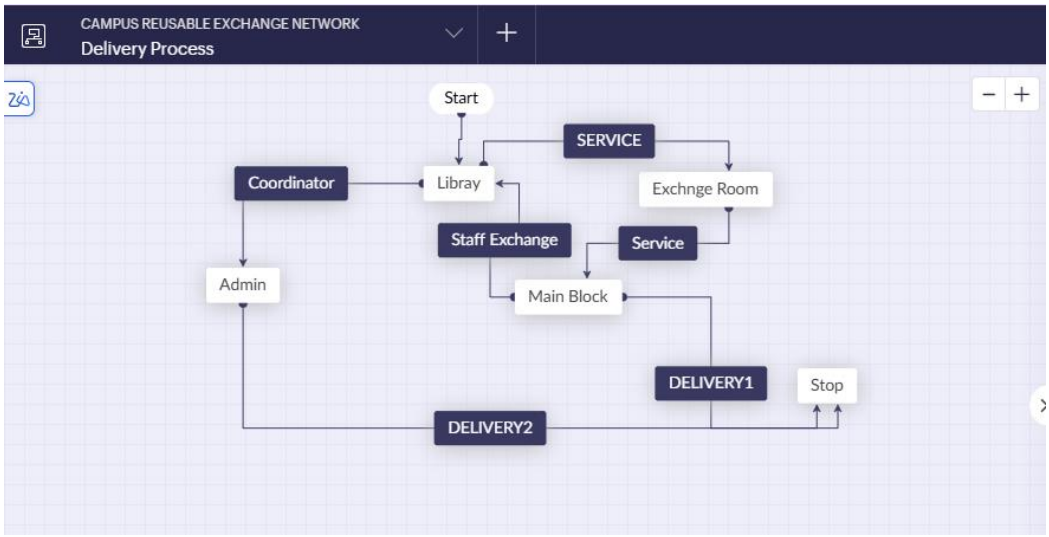


Figure 5.17: Delivery Process Blueprint Configuration

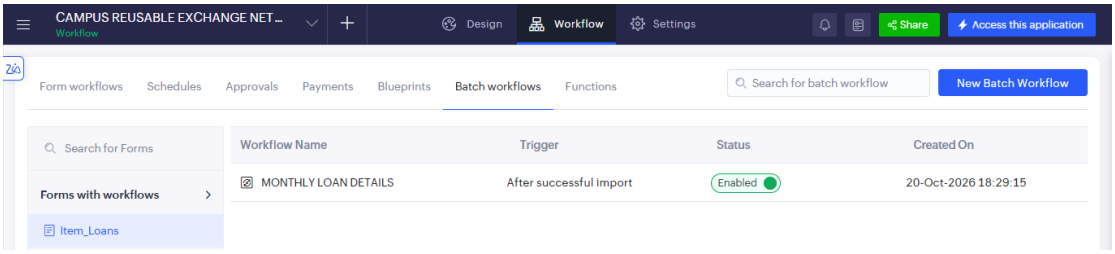


Figure 5.18: Batch Workflows List

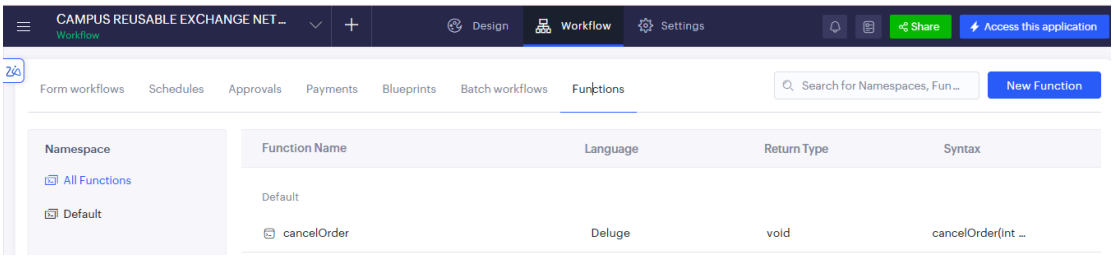


Figure 5.19: Functions List

As seen in Figure 5.14, both schedules were in the Disabled state when the screenshots were captured; they are enabled before production use (Section 7.1).

5.7 Integrations

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Invoice integration	Zoho Invoice – Invoices module (organisation CRES)	Built-in connection and integration field "Invoices" in Reusable_Items	Zoho OAuth (system connection, status Connected)	Invoice records linked to items
E-mail notification	Zoho mail via sendmail	Deluge sendmail task	Account admin user	Approval and stock notifications

The Zoho Invoice connection is authorised with the user's Zoho account and shows the status Connected with the datasource Zoho Invoice Invoices (module Invoice). The connection is referenced by the Reusable_Items form, which carries the Invoices field. No external REST calls (invokeurl) or webhooks were required.

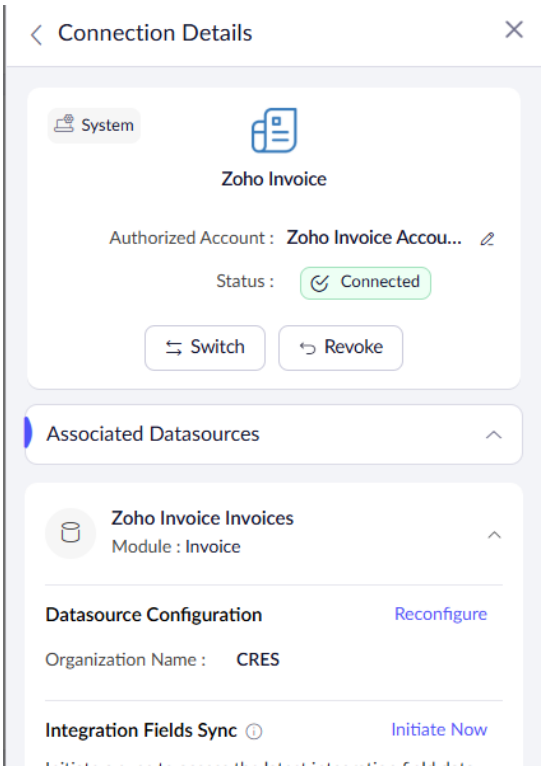


Figure 5.20: Zoho Invoice Connection Details

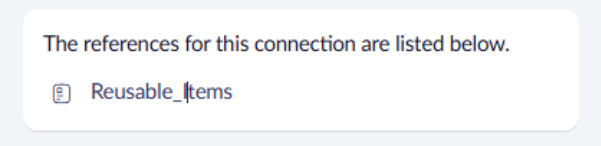


Figure 5.21: Connection Reference – Reusable_Items Form

5.8 User Management and Role-Based Access

Users are added to the application and mapped to roles (Administrator, Faculty, Staff, Student) with permission profiles that follow Table 4.11. The Role field in Campus_Users mirrors these categories and drives the Users by Role chart.

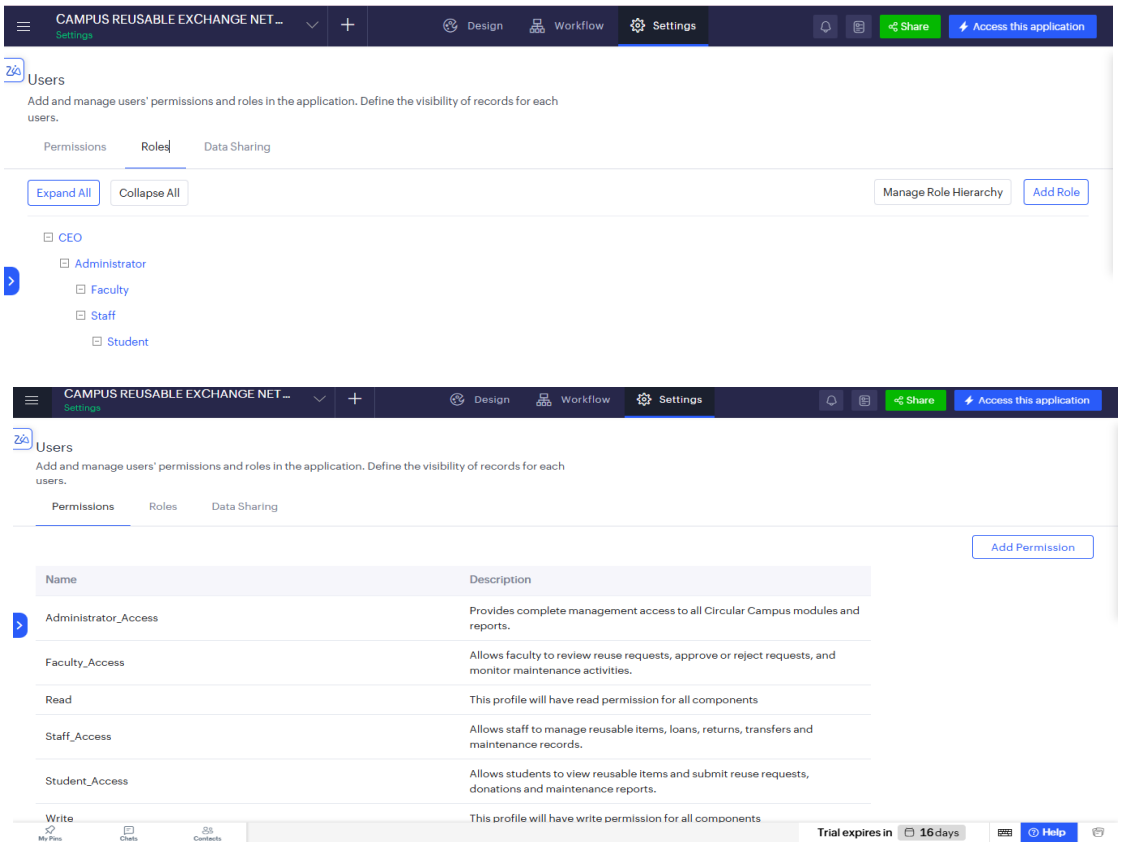


Figure 5.22: Roles and Permission Profiles Configuration

5.9 AI and Analytics Features

AI Field is used in four places: Reusable_Items has AI_Category_Prediction, AI_Keywords and Object Detection Verification fields (Figure 5.4); Maintenance_Requests has an AI Maintenance_Recommendation field (Figure 5.8); Item_Donations has a Review_Status Prediction field (Figure 5.10); and the “Circular Campus Reuse Assistant” AI agent answers

users in chat with suggestions such as “Find available reusable items”, “Submit a borrowing request”, “Register a new item donation”, “Check my return deadlines” and “Report a damaged item”. AI outputs are advisory and are checked by the reviewer before records are accepted; the agent itself displays a reminder to verify AI-generated responses. Analytics are provided by the Dashboard charts described in Section 5.4.

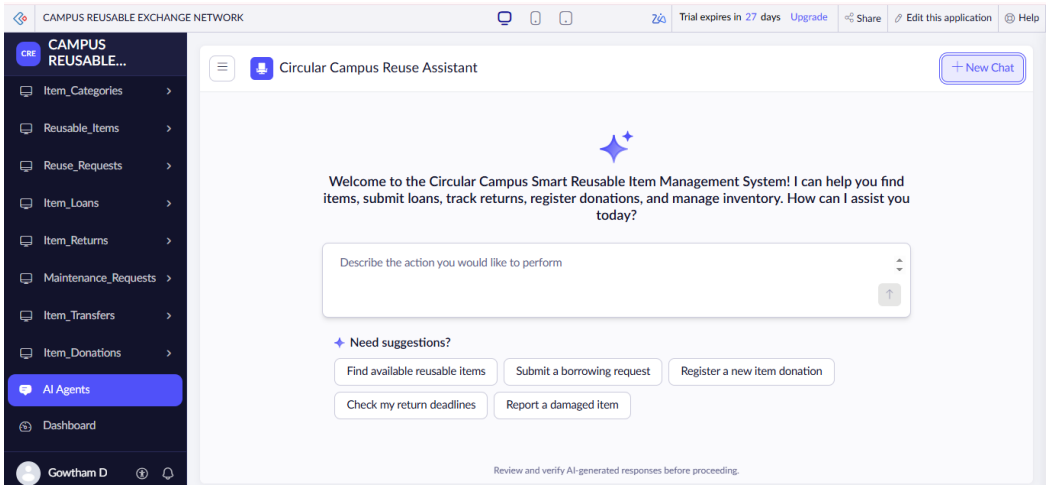


Figure 5.23: Circular Campus Reuse Assistant (AI Agent)

5.10 Pro-Code Extension

No pro-code extension (Zoho Catalyst, JavaScript widgets or Creator REST API clients) was used, because every requirement could be met with forms, workflows and Deluge. If heavy processing or external system integration is needed later, a Catalyst function could be invoked from Deluge, and a widget could replace the dashboard for custom visualisation.

5.11 Mobile and Portal Access

The application is built with the responsive layout and can be previewed in desktop, tablet and mobile modes from the builder toolbar; the same forms are available in the Zoho Creator mobile app. No customer portal was configured.

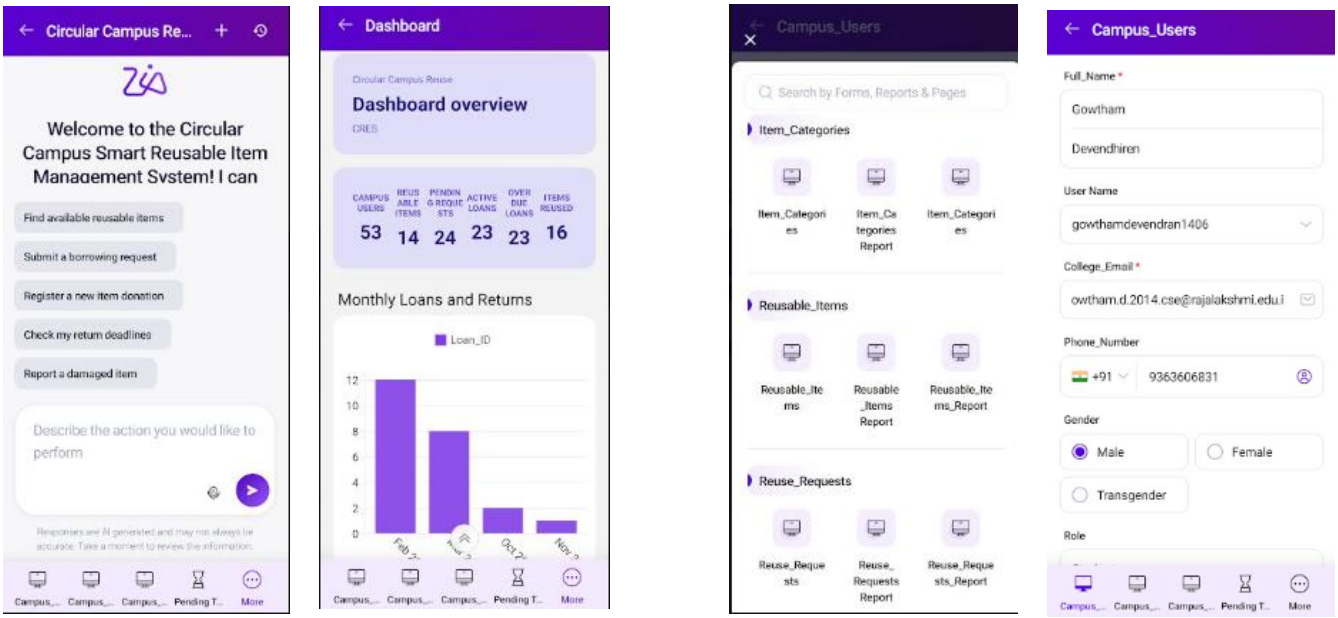


Figure 5.24: Application on Mobile

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out at four levels: unit testing of each form (mandatory fields, lookups and uploads), script testing of every Deluge script with info statements, integration testing of the request → approval → loan → return flow and the Zoho Invoice connection, and role-based testing by logging in as each role. User acceptance testing followed with real users.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Expected Result	Actual	Status
TC-01	FR-01	Submit Campus_Users with all mandatory fields	Record created	As Expected	PASS
TC-02	FR - 14	User With Same Reg Number	Error Occurs	As Expected	PASS
TC-02	FR-01	Submit Campus_Users with College_Email blank	Mandatory error shown	As Expectd	PASS
TC-03	FR-02	Add Reusable_Items with category and image	Item saved; AI fields populated	As Expectd	PASS
TC-04	FR-03	Submit Reuse_Requests with invalid data	Not Submit	As Expectd	FAIL
TC-05	FR-03	Submit valid Reuse_Requests	Request saved	As Expectd	PASS
TC-06	FR-04	Create request	Approver receives e-mail	As Expectd	PASS
TC-07	FR-05	Approve request	Status set Approved	As Expectd	PASS
TC-08	FR-05	Reject request	Status set Rejected	As Expectd	PASS
TC-09	FR-06	Issue loan without Due_Date	Mandatory error shown	As Expectd	PASS
TC-10	FR-07	Record return with damage charge	Return saved	As Expectd	PASS
TC-11	FR-08	Create maintenance request	Approver e-mailed (DS-01)	As Expectd	PASS
TC-12	FR-09	Set status to In Progress	Item set to Maintenance	As Expectd	PASS
TC-13	FR-10	Create transfer	Transfer saved with status	As Expectd	PASS
TC-14	FR-11	Submit donation for review	Item Approval routed to approver	As Expectd	PASS
TC-15	FR-12	Open Dashboard	KPIs and charts shown	As Expectd	PASS
TC-16	FR-13	Ask AI agent to find items	Relevant reply shown	As Expectd	PASS
TC-17	NFR-02	Student opens Administrator-only report	Access denied	As Expectd	PASS
TC-18	FR-03	cancelOrder on an Approved request	“can no longer be cancelled” message	As Expectd	PASS

TC ID	FR Ref.	Test Description	Expected Result	Actual	Status
TC – 19	FR – 15	After Approved It moves from Pending To Completed	Product List Is Transferred	Not As Expected	FAIL

The Actual and Status columns are completed after executing the cases on the live application.

6.3 Debugging Deluge Scripts

Scripts were debugged using info statements (for example "Approver not found or has no email address…" in DS-01), the Deluge editor, and workflow execution logs. A real defect was found in DS-02: the else-if branch refers to maintenanceRecord, which is only fetched in the Completed branch, so the lookup is empty when the status is In Progress or Assigned. The fix is to fetch maintenanceRecord before the if statement.

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Item status not set to Maintenance on In Progress / Assigned	maintenanceRecord fetched only inside the Completed branch	Fetch record at the top of the script	Closed
BUG-02	Daily_Overdue_Check does not update loans	Loop body contains only comments	Add status update and reminder statements	Open
BUG-03	Schedules not running	Schedules in Disabled state	Enable before production	Closed

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Haripraveen N	Student	The application is easy to use. Reusable items can be viewed and reuse requests can be submitted successfully.	Improved the form layout and verified the reuse request submission workflow.
Gowtham D	Administrator	User management, approval workflows and maintenance tracking are useful. Some workflow notifications required testing.	Configured approval workflows, email notifications and Deluge validation scripts.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

The application was developed in the Development environment, tested with sample data (50 users and 13 items as seen on the Dashboard) and published for use through the Access this application link. Before production the schedules are enabled, test data is isolated, connections are re-authorised and version notes are recorded.

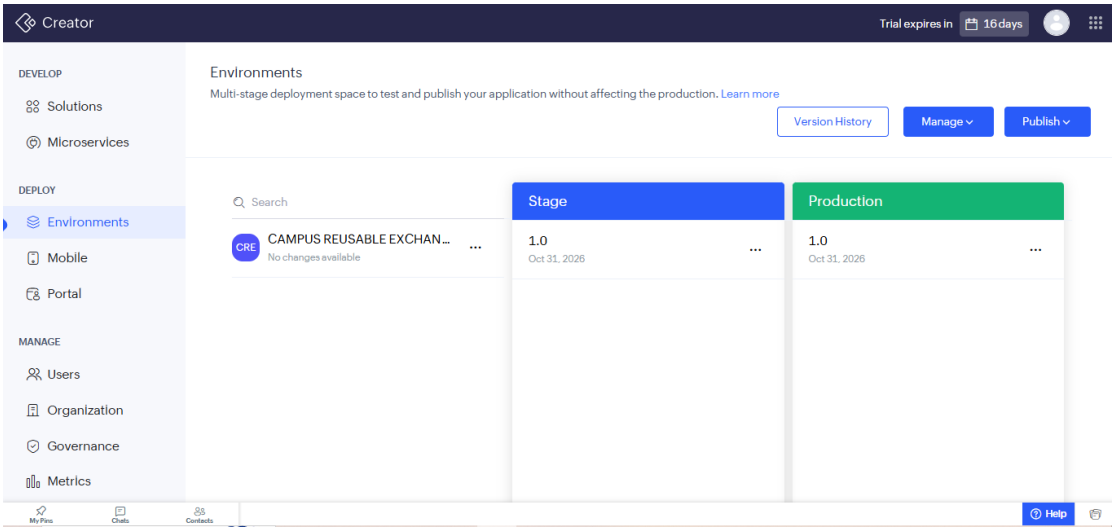


Figure 7.1: Environments / Publish to Production

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓
2	Sample/test data removed or isolated	✓
3	Roles, permissions and users configured in Production	✓
4	Connections authorised in Production (Zoho Invoice)	✓
5	Schedules and workflows enabled	✓
6	Application published and URL verified	✓
7	Version / release notes recorded	✓

7.2 Performance Optimisation

Scripts use criteria-based fetches (for example Item_Loans[Loan_Status == "Active" && Due_Date < todayDate]) instead of scanning the whole form, avoid fetching inside loops, return early in cancelOrder, and the batch workflow processes imported loans five records at a time.

Table 7.2: Performance Optimisation Applied

Optimisation Applied	Component	Before	After	Benefit
Early return on invalid state	DS-03	Full processing	Exit on first failed check	Less work per call
Batch size of 5 records	MONTHLY LOAN DETAILS	Single large run	Controlled batches	Safer imports

7.3 Security Features

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role-based permissions	Forms, reports and page	Unauthorised data access
Mandatory fields and validation	Forms, Validate_Reuse_Request	Invalid or incomplete data
OAuth connection (no hard-coded keys)	Zoho Invoice	Credential leakage
Approval before action	Item_Donations, Reuse_Requests	Unapproved transactions
Signature capture	Campus_Users	Repudiation

7.4 Governance and Administration

Naming follows form-based link names (for example Reusable_Items), workflows carry descriptive names, and the audit trail records additions and changes. Data can be exported from reports for backup, and the application owner (the administrator) is responsible for user licences and maintenance.

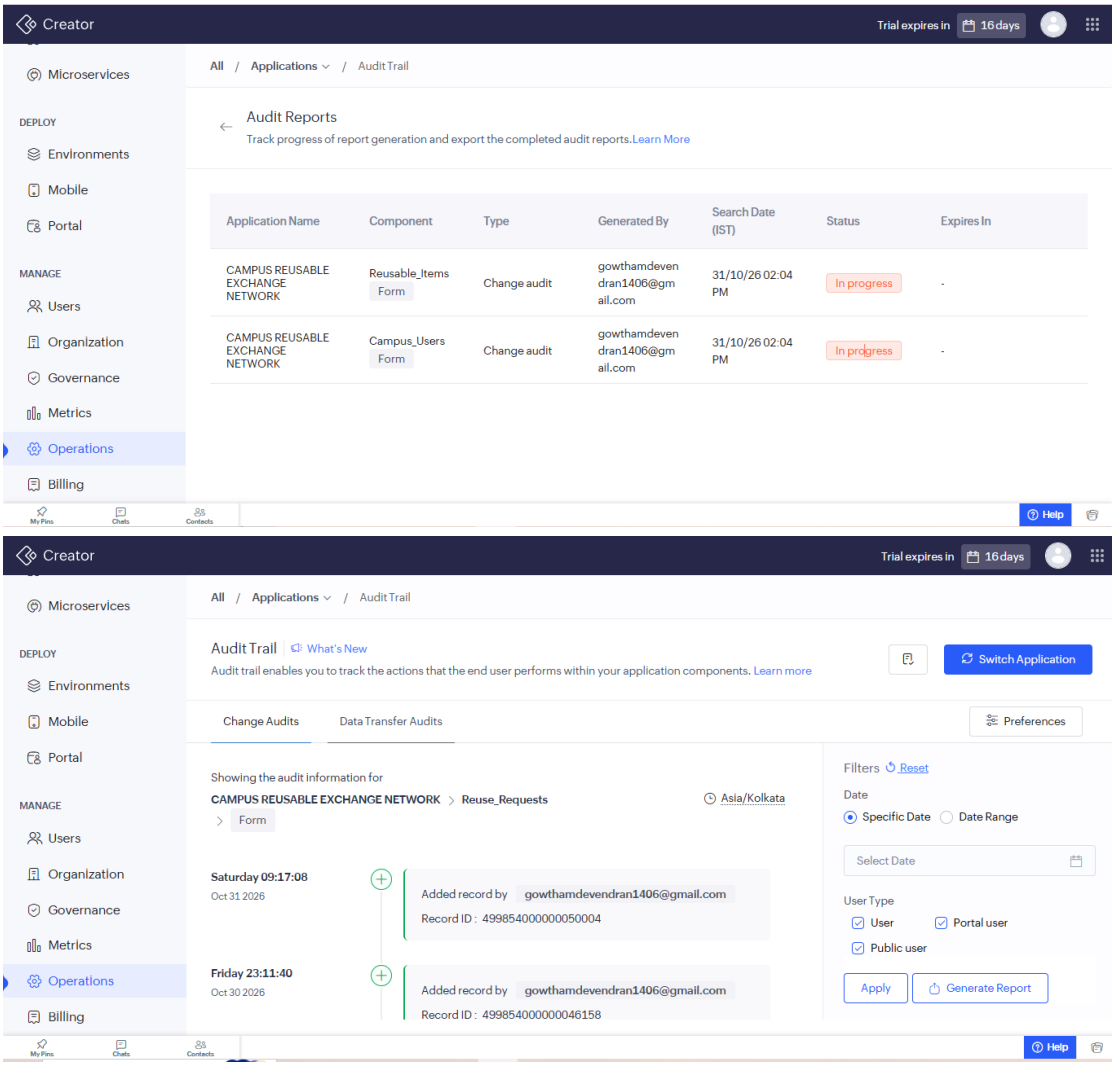
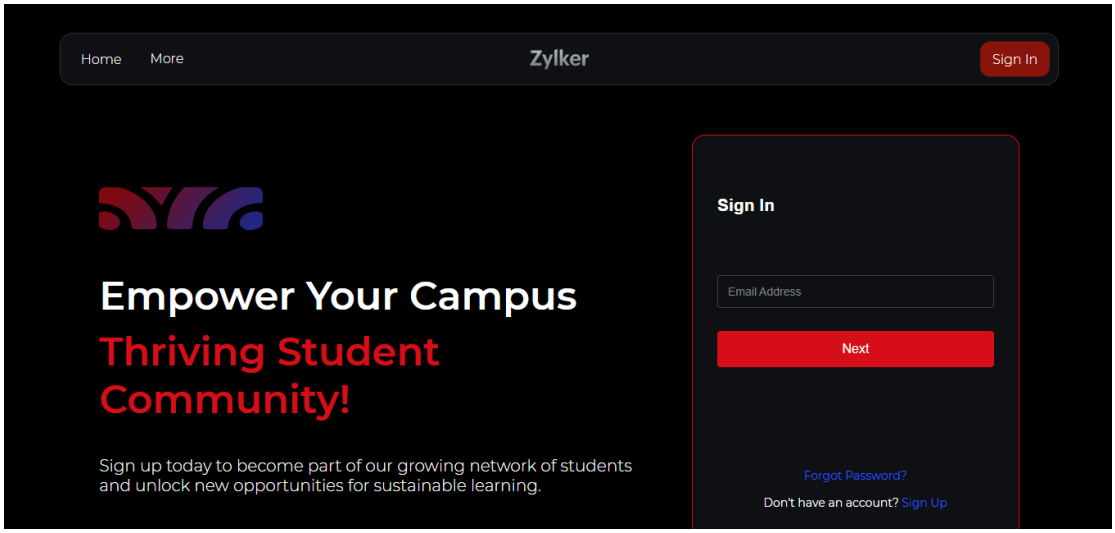


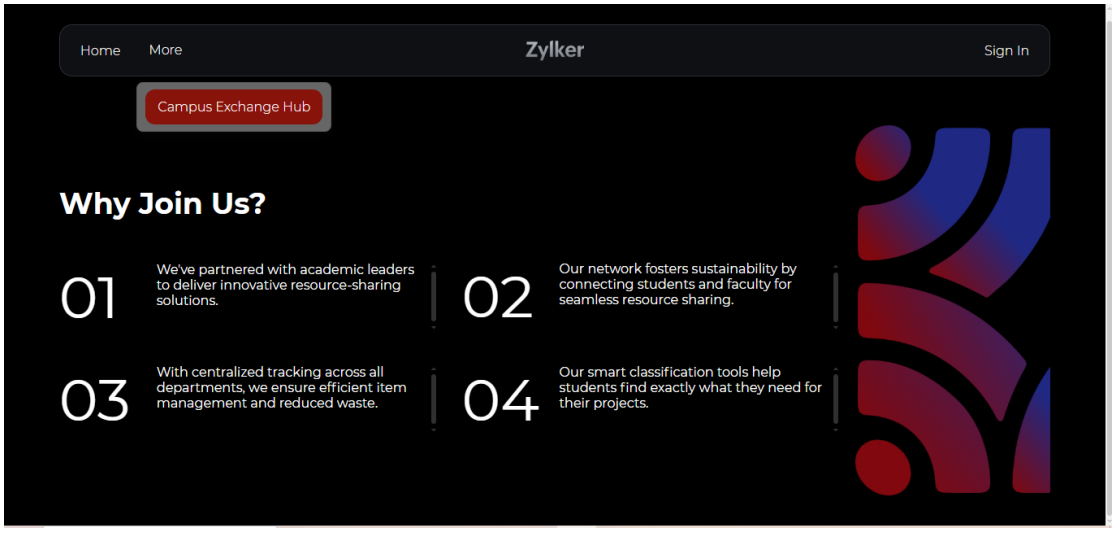
Figure 7.2: Audit Trail / Administration Settings

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

A typical scenario runs as follows. A student submits a request; the approver is notified and approves; the Dashboard counters update.





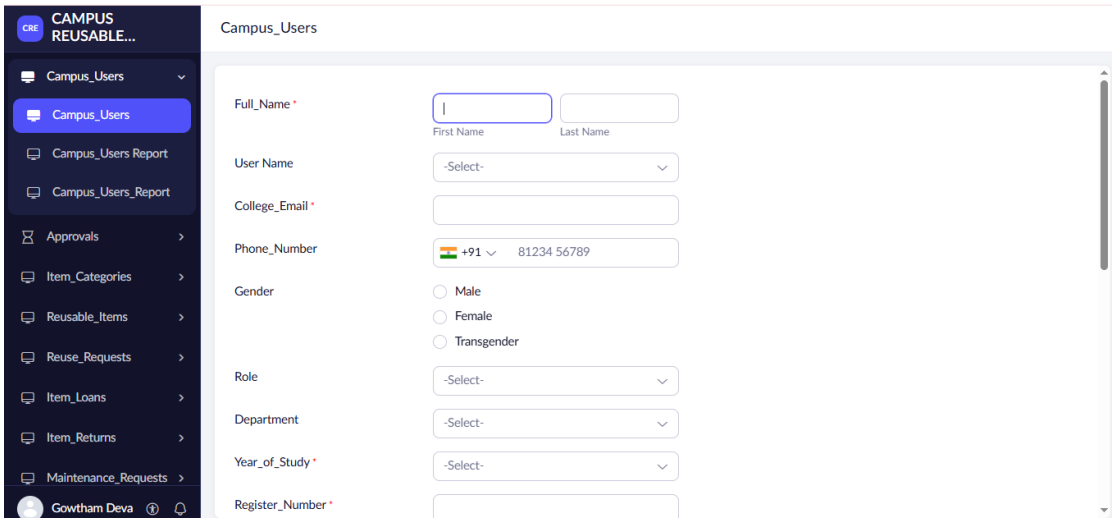
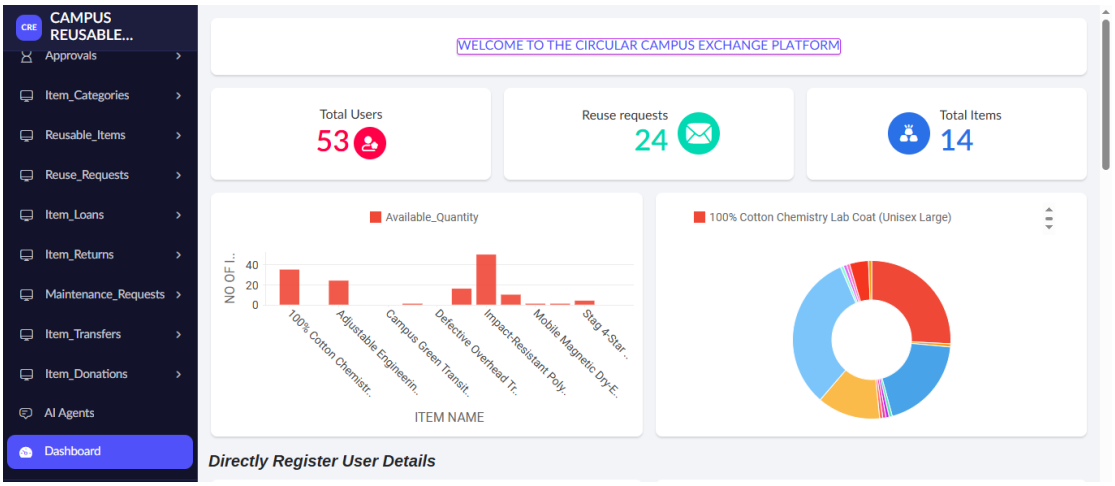


Figure 8.1: Live URL



8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Data model and forms	Section 5.2	Nine forms with lookups
Automation	Sections 5.5, 5.6	Form, schedule, approval, Blueprint, batch workflows and Deluge scripts
Dashboards and reports	Sections 5.3, 5.4	18 reports and Dashboard page
Integration and AI	Sections 5.7, 5.9	Zoho Invoice connection, Zia AI fields and AI agent
Secure deployment	Chapter 7	Roles, approvals and OAuth connection

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Item visibility	None; asked by word of mouth	Searchable catalogue	Real-time availability
Approval	Verbal	Recorded approval process	Full audit trail
Overdue tracking	Memory	Daily schedule	Automatic detection
Reports	Manual	Dashboard and reports	Instant
Damage handling	Unrecorded	Return inspection and maintenance	Accountability

8.4 Limitations of the Low-Code Approach

The main constraints met were limits on visual customisation of forms and dashboards, dependence on Zoho's platform features and trial edition, and the need to write Deluge for non-trivial logic such as cancellation and completion handling. Vendor lock-in is also a consideration. Workarounds were to use a page with HTML snippets for the dashboard layout and custom functions for reusable logic.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Forms, reports, the Dashboard, approvals and Blueprint were configured without code (no-code). Notification, status updates, cancellation and overdue checks needed Deluge (low-code). Barcode scanning, payment collection or advanced analytics would move the project to pro-code using Catalyst or external services.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

The Campus Reusable Exchange Network provides a working, role-based platform for managing the full life cycle of reusable campus items: registration, cataloguing, request, approval, loan, return, maintenance, transfer and donation. Workflows, Deluge scripts, Zia AI and a Zoho Invoice connection automate routine work, and the Dashboard gives managers an instant view of users, items, requests and loans.

The project shows that a complete business application can be delivered quickly on a low-code platform and be maintained by non-programmers, while leaving a clear path to pro-code where needed.

9.2 Future Enhancements

- Complete the Daily_Overdue_Check script to update status, send reminders and apply fines.
- Add QR-code scanning for issue and return.
- Enable a portal for external donors.
- Use Zia predictions for demand forecasting and a Catalyst function for heavy processing.
- Add Kanban and Calendar reports for requests and due dates.

9.3 Learning Outcomes and Career Relevance

The project built skills in data modelling, Deluge scripting, workflow and approval design, AI features, integration and governance on Zoho Creator, which relate directly to roles such as citizen developer, low-code developer, solutions consultant and automation engineer.

REFERENCES

[1] W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide. New Delhi, India: BPB Publications.

[2] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/ (Accessed: 31/10/2026).

[3] Zoho Corporation, “Deluge Script Reference Guide.” [Online]. Available: https://www.zoho.com/deluge/ (Accessed: 31/10/2026).

[4] Zoho Corporation, “Zoho Creator Learning Centre.” [Online]. Available: https://www.zoho.com/creator/resources/ (Accessed: 31/10/2026).

[5] Ellen MacArthur Foundation, “Towards the Circular Economy, Vol. 1,” 2013.

[6] A. C. Bock and U. Frank, “Low-Code Platform,” Business & Information Systems Engineering, vol. 63, pp. 733–740, 2021.

[7] A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, “Supporting the understanding and comparison of low-code development platforms,” in Proc. 46th Euromicro SEAA, 2020, pp. 171–178.

[8] M. Waszkowski, “Low-code platform for automating business processes in manufacturing,” IFAC-PapersOnLine, vol. 52, no. 10, pp. 376–381, 2019.

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Sec.
I	Application created & sections/menu organised	Y	CRE application navigation	5.1
II	Forms with ≥ 8 distinct field types	Y	All nine forms	5.2
II	Lookup field	Y	Reusable_Items, Reuse_Requests, etc.	4.2, 5.2
II	Subform (parent–child data)	Y	Item_Loans → Item Return	4.2, 5.2
II	Field validations / layout rules	Y	Mandatory fields; Validate_Reuse_Request	5.2
II	List report with filter, grouping, custom action	Y (list reports)	Reports per form	5.3
II	Kanban / Calendar / Timeline / Map report	Y	Used in Different Forms	5.3
II	Pivot table / chart report	Y (charts)	Dashboard charts	5.3, 5.4
II	Page / dashboard built with UI Builder	Y	Dashboard	5.4
II	Deluge – form events	Y	DS-01, DS-05	5.5
II	Deluge – functions, fetch, loops, collections	Y	DS-02, DS-03, DS-04	5.5
II	Email / in-app notifications	Y	Maintainence Approval	5.5, 5.6
II	Form workflow	Y	Six form workflows	5.6
II	Scheduled workflow	Y	Monthly Loan Detils ,DAILY STOCK DETAILS	5.6
II	Approval process	Y	Item Approval	5.6
II	Blueprint	Y	Delivery Process	5.6
II	Batch / report workflow	Y	MONTHLY LOAN DETAILS	5.6
II	Testing & debugging evidence	Y	Chapter 6	6
III	Integration with a Zoho service	Y	Zoho Invoice	5.7
III	Connection / REST API call (invokeurl)	N	—	5.7
III	Webhook or Creator REST API	N	—	5.7
III	Users, roles and permission profiles	Y	Administrator , Staff , Faculty & Student	5.8
III	Record-level access criteria / customer portal	Y	Administrator , Staff , Faculty & Student	5.8
III	Zia / AI feature	Y	AI fields and AI agent	5.9
III	Analytics (charts / Zoho Analytics)	Y	Dashboard charts	5.9

Unit	Zoho Creator Feature	Done	Where in the Application	Sec.
IV	Performance optimisation applied & measured	Y (qualitative)	DS-04, batch	7.2
IV	Deployment across environments	Y	Version 1	7.1
IV	Security controls	Y	Zoho Invoice OAuth	7.3
IV	Governance (audit trail, administration)	Y	Reuse Request Form	7.4
V	Pro-code extension	N	Not used	5.10
V	Low-code limitations & spectrum analysis	Y	Sections 8.4, 8.5	8.4, 8.5
—	Mobile app / portal access	Y	https://gowthamdevendran1406.zohocreatorportal.in	5.11

Total features implemented: 29 / 32 (items marked [Verify] are confirmed against the live application before submission).

APPENDIX B – COURSE OUTCOME MAPPING

Self-assess the attainment Level (1 - Low , 2 – Medium , 3 – High)

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	3	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	2	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	1	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Maintenance_Requests → Maintenance_Approval_Request

```
sendmail
[
	from :zoho.adminuserid
	to :"gowthamdevendran1406@gmail.com"
	subject :"New Maintenance Approval Request - Circular Campus"
	message :"A new maintenance request has been submitted in Circular Campus.\n\n" + "Maintenance ID: " + input.Maintenance_ID + "\n" + "Issue Type: " + input.Issue_Type + "\n" + "Priority: " + input.Priority + "\n" + "Status: " + input.Maintenance_Status + "\n\n" + "Please log in to Circular Campus and review the maintenance request."
]
```

C.2 Function cancelOrder

```
void cancelOrder(int requestID, string reason)
{
	// Find request using Request_ID
	req = Reuse_Requests[Request_ID == requestID];
	if(req.ID == null)
	{
		info "Request not found.";
		return;
	}
	// Check request status
	if(req.Request_Status == "Cancelled")
	{
		info "Request is already cancelled.";
		return;
	}
	if(req.Request_Status == "Approved" || req.Request_Status == "Completed")
	{
		info "This request cannot be cancelled.";
		return;
	}
	// Find the linked Campus_Users record
	userRecord = Campus_Users[ID == req.Full_Name];
	if(userRecord.ID == null)
	{
		info "Linked Campus User not found.";
		return;
	}
	// Get registered email
	userEmail = userRecord.College_Email;
	if(userEmail == null || userEmail == "")
	{
		info "User email address is missing.";
```

```
		return;
	}
	// Set cancellation reason
	if(reason == null || reason.trim() == "")
	{
		reason = "No reason provided";
	}
	// Update request status
	req.Request_Status="Cancelled";
	req.Approval_Notes="Cancelled: " + reason;
	// Send email to the selected user
	sendmail
	[
		from :zoho.adminuserid
		to :userEmail
		subject :"Circular Campus - Reuse Request Cancelled"
		message :"Hello " + userRecord.Full_Name + ",\n\nYour reuse
request has been cancelled successfully.\n\nRequest ID: " + req.Request_ID +
"\nQuantity: " + req.Quantity_Requested + "\nReason: " + reason + "\nStatus:
Cancelled\n\nThank you,\nCircular Campus Team"
	]
	info "Request cancelled successfully.";
	info "Email sent to: " + userEmail;
}
```

C.3 Reuse Item Request Approval

```
requestRecord = Reuse_Requests[ID == input.ID];
if(requestRecord != null)
{
	requestRecord.Request_Status="Approved";
	requestRecord.Approver=zoho.loginuser;
	requestRecord.Approval_Notes="Request approved by " + zoho.loginuser;
	info "Request approved successfully.";
}
```

C.4 Schedule Monthly Loan Details

```
previousMonthDate = zoho.currentdate.toStartOfMonth().addMonth(-1);
reportMonth = previousMonthDate.toString("MMMM-yyyy");
reportMonthNumber = previousMonthDate.getMonth();
reportYear = previousMonthDate.getYear();
totalLoans = 0;
returnedLoans = 0;
pendingLoans = 0;
overdueLoans = 0;
totalFines = 0.0;
for each  loan in Item_Loans
{
```

```
	if(loan.Issue_Date != null)
	{
		if(loan.Issue_Date.getMonth() == reportMonthNumber &&
loan.Issue_Date.getYear() == reportYear)
		{
			totalLoans = totalLoans + 1;
			if(loan.Loan_Status == "Returned")
			{
				returnedLoans = returnedLoans + 1;
			}
			else if(loan.Loan_Status == "Pending")
			{
				pendingLoans = pendingLoans + 1;
			}
			else if(loan.Loan_Status == "Overdue")
			{
				overdueLoans = overdueLoans + 1;
			}
			if(loan.Fine_Amount != null)
			{
				totalFines = totalFines + loan.Fine_Amount;
			}
		}
	}
}
emailMessage = "CIRCULAR CAMPUS - MONTHLY LOAN REPORT\n\n";
emailMessage = emailMessage + "Report Month: " + reportMonth + "\n\n";
emailMessage = emailMessage + "TOTAL LOAN DETAILS\n";
emailMessage = emailMessage + "Total Loans: " + totalLoans + "\n";
emailMessage = emailMessage + "Returned Loans: " + returnedLoans + "\n";
emailMessage = emailMessage + "Pending Loans: " + pendingLoans + "\n";
emailMessage = emailMessage + "Overdue Loans: " + overdueLoans + "\n";
emailMessage = emailMessage + "Total Fine Amount: Rs. " + totalFines + "\n\n";
emailMessage = emailMessage + "This is the automatically generated monthly
report from Circular Campus.";
sendmail
[
	from :zoho.adminuserid
	to :"gowtham.d.2024.cse@rajalakshmi.edu.in"
	subject :"Circular Campus - Monthly Loan Report - " + reportMonth
	message :emailMessage
]
info "Monthly loan report email sent.";
info emailMessage;
```

C.5 BATCH WORKFLOW MONTHLY LOAN DETAILS

```
loanCount = 1;
returnedCount = 0;
pendingCount = 0;
overdueCount = 0;
fineAmount = 0.0;
if(input.Loan_Status == "Returned")
{
        returnedCount = 1;
}
else if(input.Loan_Status == "Pending")
{
        pendingCount = 1;
}
else if(input.Loan_Status == "Overdue")
{
        overdueCount = 1;
}
if(input.Fine_Amount != null)
{
        fineAmount = input.Fine_Amount;
}
info "Monthly Loan Record Processed";
info "Loan ID: " + input.ID;
info "Total Loans: " + loanCount;
info "Returned: " + returnedCount;
info "Pending: " + pendingCount;
info "Overdue: " + overdueCount;
info "Fine: " + fineAmount;
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the application link https://gowthamdevendran1406.zohocreatorportal.in in a browser and sign in with your Zoho account, or open the Zoho Creator mobile app and select Campus Reusable Exchange Network.

D.2 Instructions for Student / Faculty

- Open Campus_Users and submit your details (once).
- Open Reuse_Requests, select the item, quantity, dates and purpose, and press Submit.
- Use AI Agents to find items or report a damaged item.
- Donate an item through Item_Donations.

D.3 Instructions for Administrator / Staff

- Review new requests and approve or reject them.
- Issue approved items through Item_Loans and record returns in Item_Returns.
- Handle Maintenance_Requests and Item_Transfers.
- Open Dashboard and the reports to monitor loans, overdue items and latest updates.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	GOWTHAM D	Register No.	2116240701154
Project Title	CAMPUS REUSABLE EXCHANGE NETWORK	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		100	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner Name: _____	Signature of External Examiner Name: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	GOWTHAM D	Register No.	2116240701154
Department / Section	CSE	Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	100	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER Name: _____	EXTERNAL EXAMINER Name: _____